Advanced Concepts in Machine Learning for Energy

Time: March 26, 27, 30, April 1, 3

1:00 pm - 5:30 pm (in-person students)

(10 more days for distance students)

Location: GERB 315 and Zoom

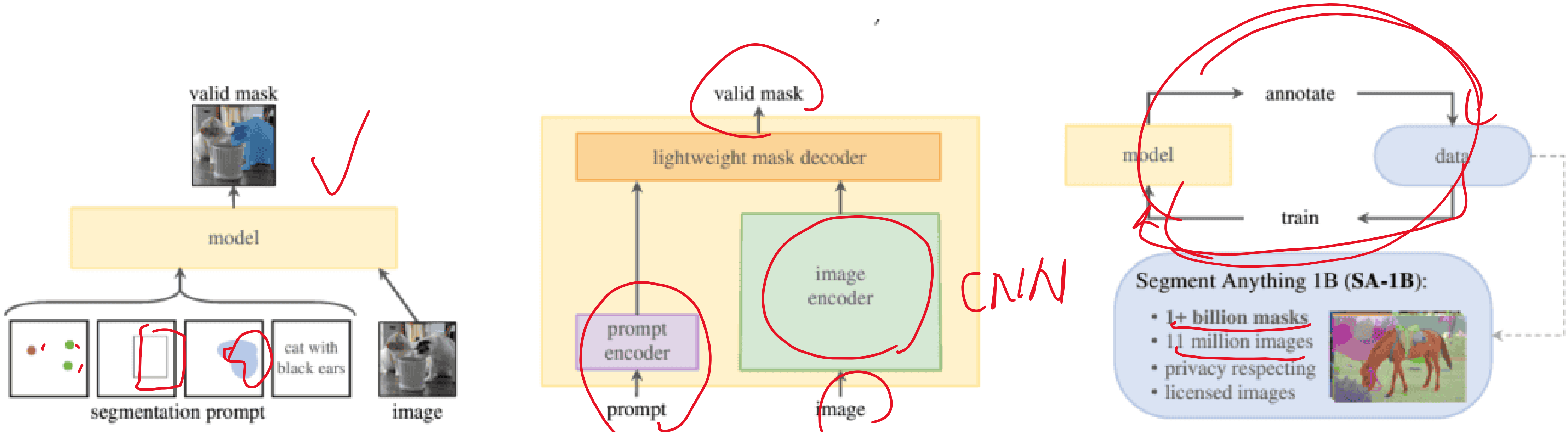
Lecture	Topic	Quiz
1	Neural network fundamentals and continuous optimization	
2	Image classification, object detection, and pixel segmentation	#1
3	Autoencoder, autodecoder, and generative adversarial network (GAN)	#2 generative AI
4	Attention mechanism, transformers (GPT)	#3
5	Final project Q&A, advanced topics review	#4

Lecture 3

- Quiz #2 on Canvas (1:00-1:30 pm)
- Advanced computer vision (1:30-2:00 pm)
- Lab 3a (2:00-2:30 pm)
- Generative AI (2:30-3:10 pm)
- Lab 3b(3:10-3:40 pm)
- Anomaly Detection using Autoencoders (3:40-4:00 pm)
- **Lab 3c** (4:00-4:30 pm)
- GAN(4:30-5:30 pm)
- **Lab 3d** (5:00-5:30 pm)

Segment anything model (SAM)

- Foundational model for segmentation
- https://aidemos.meta.com/segment-anything/

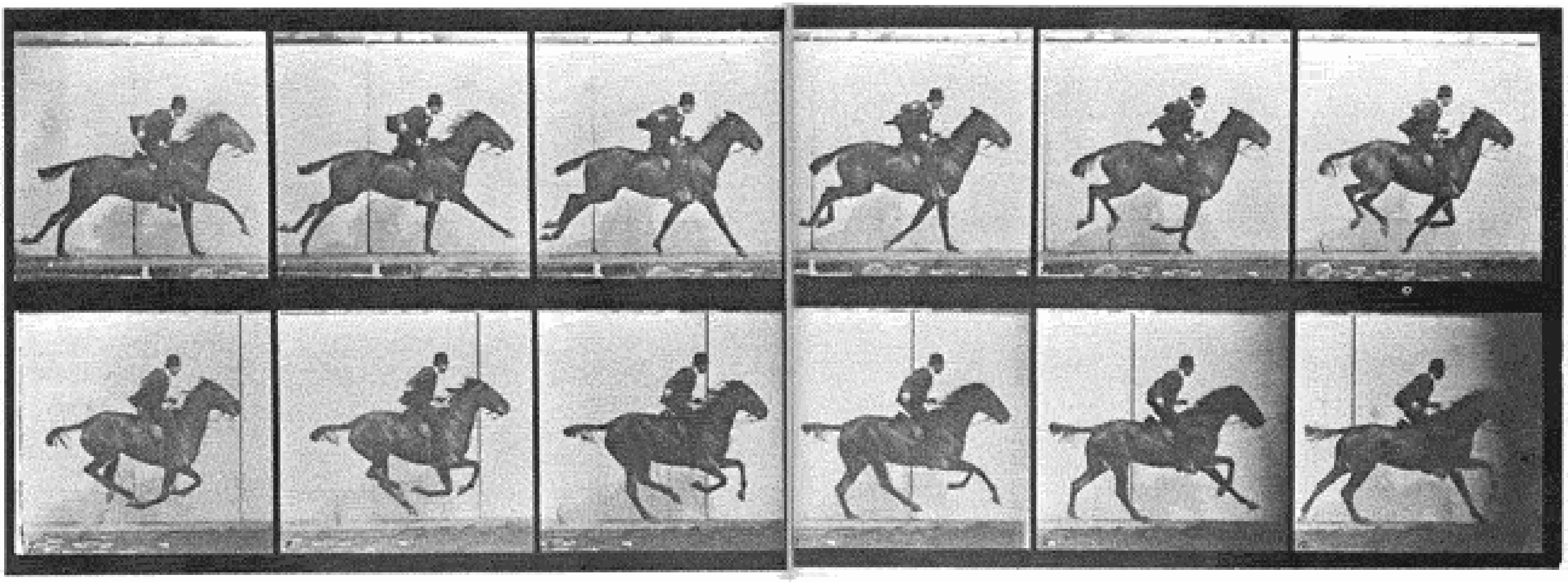


(a) **Task**: promptable segmentation (b) **Model**: Segment Anything Model (**SAM**) (c) **Data**: data engine (top) & dataset (bottom)

Segment Anything, ICCV 2023 SAM 2: Segment Anything in Images and Videos, 2024

SAM 3D body

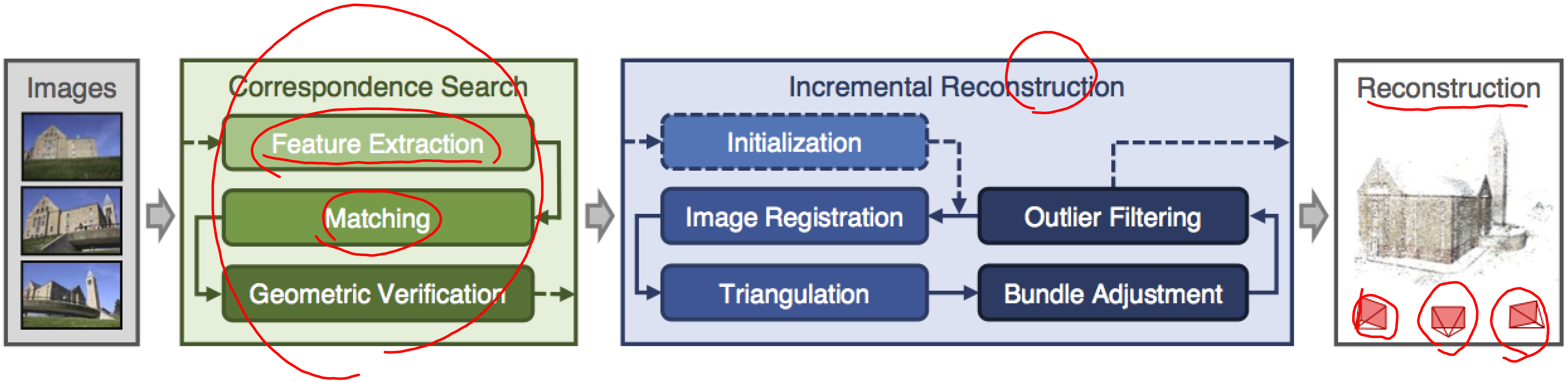
Video processing



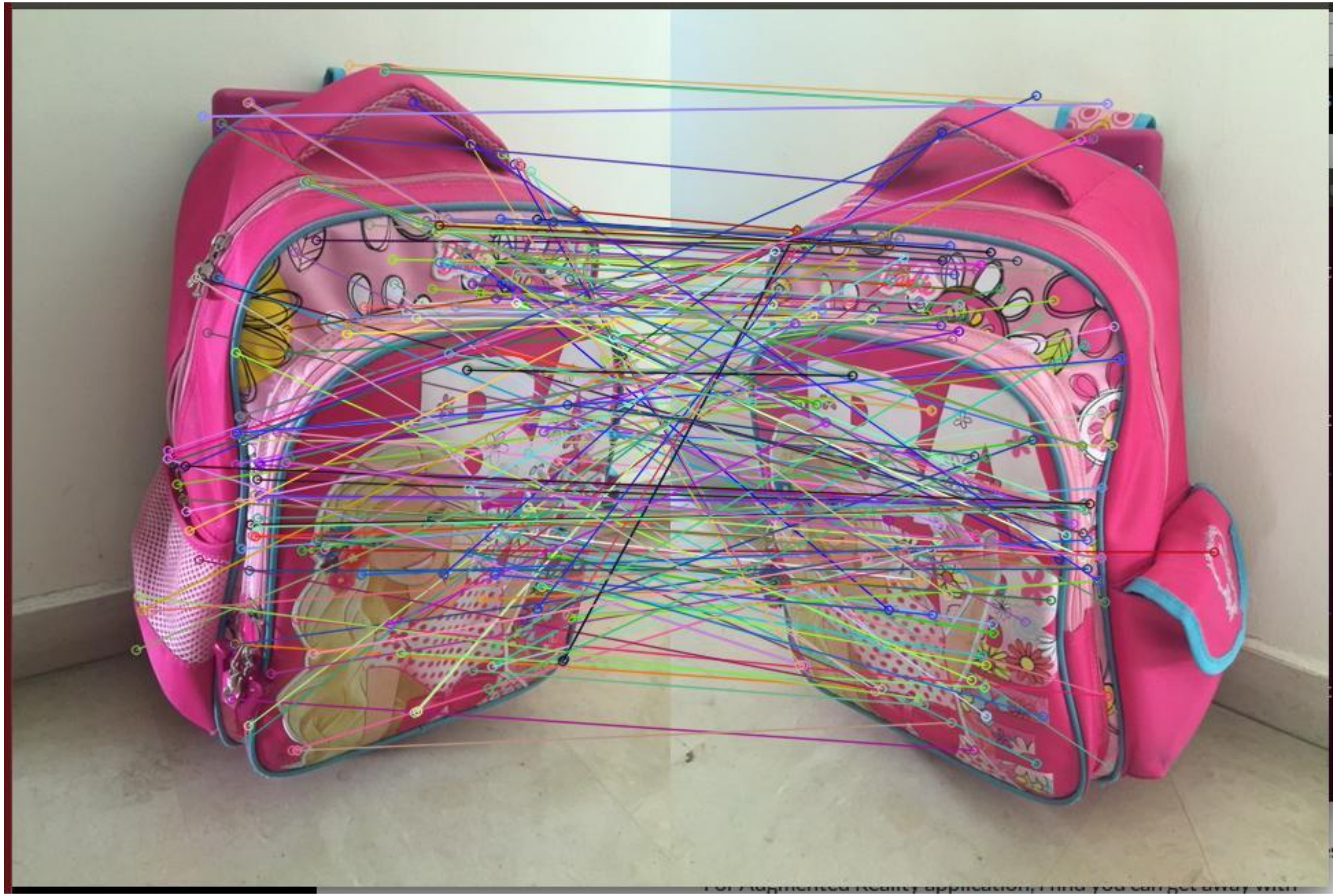
The first-ever stop-motion film was created by Eadweard Muybridge in 1878. Called *The Horse in Motion*, it’s an 11-frame clip that was created to settle a debate over whether all four hooves of a horse were off the ground at the same time when it gallops.

3D reconstruction

- COLMAP
- Scale-invariant feature transform (SIFT)



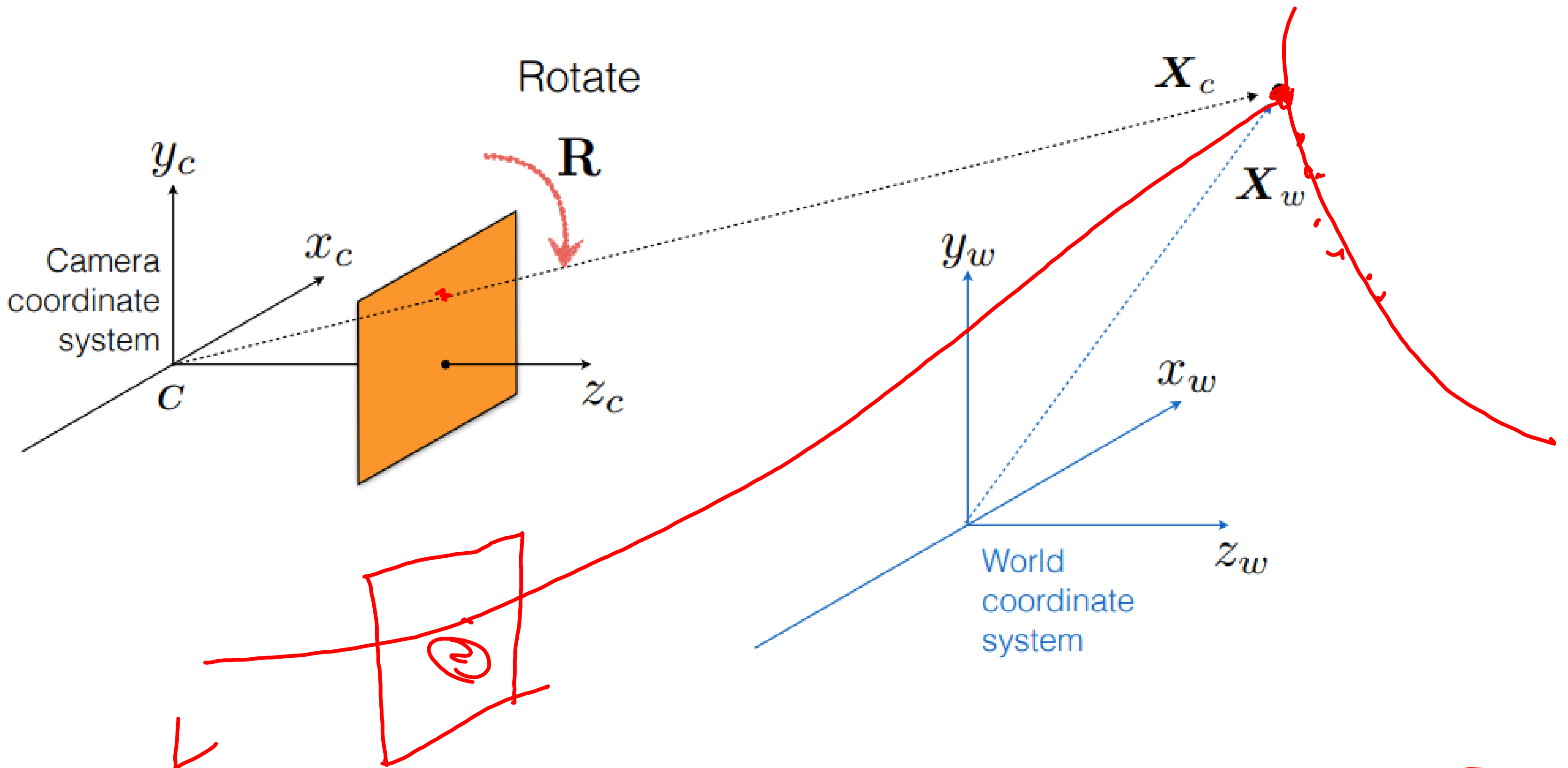
Feature matching



7-8 pairs

$\begin{bmatrix} u_1 \\ v_1 \end{bmatrix}$

$\begin{bmatrix} u_2 \\ v_2 \end{bmatrix}$



K

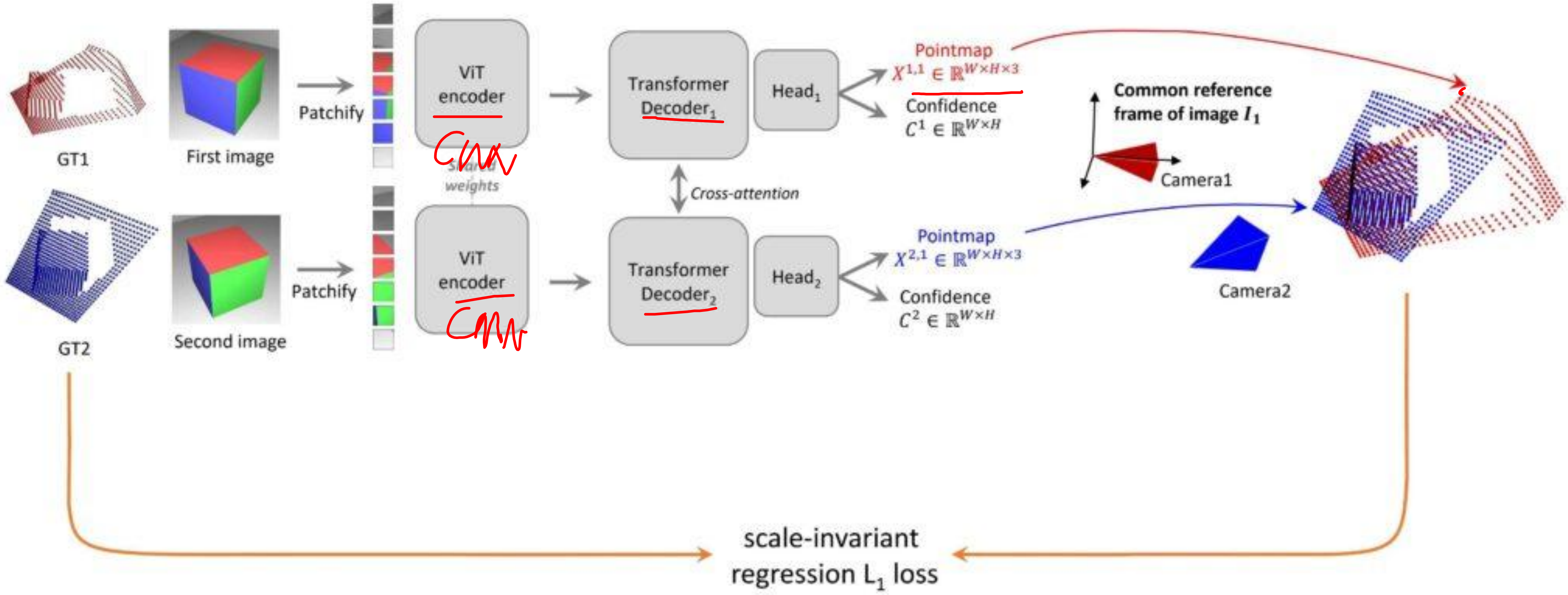
$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} a_u & 0 & u_o \\ 0 & a_v & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} r_1 & r_2 & r_3 & t_1 \\ r_4 & r_5 & r_6 & t_2 \\ r_7 & r_8 & r_9 & t_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

$$P = K[R|T]$$

Figure 3.6: *Incremental structure from motion.* My incremental structure from motion approach reconstructs the scene a few cameras at a time. This sequence of images shows the **Trevi** data set at three different stages during incremental reconstruction. Left: the initial two-frame reconstruction. Middle: an intermediate stage, after fifteen images have been added. Right: the final reconstruction with 360 photos.

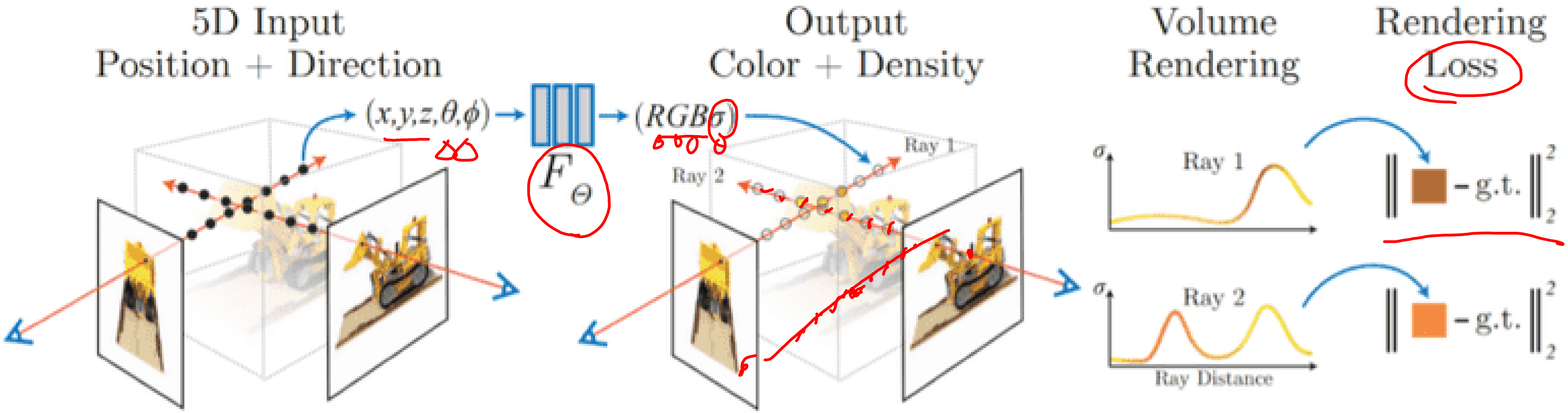
D. Nist ´er. Automatic Dense Reconstruction from Uncalibrated Video Sequences, PhD Thesis, Royal Institute of Technology KTH, ISBN 91-7283-053-0, March 2001.

Dust3R

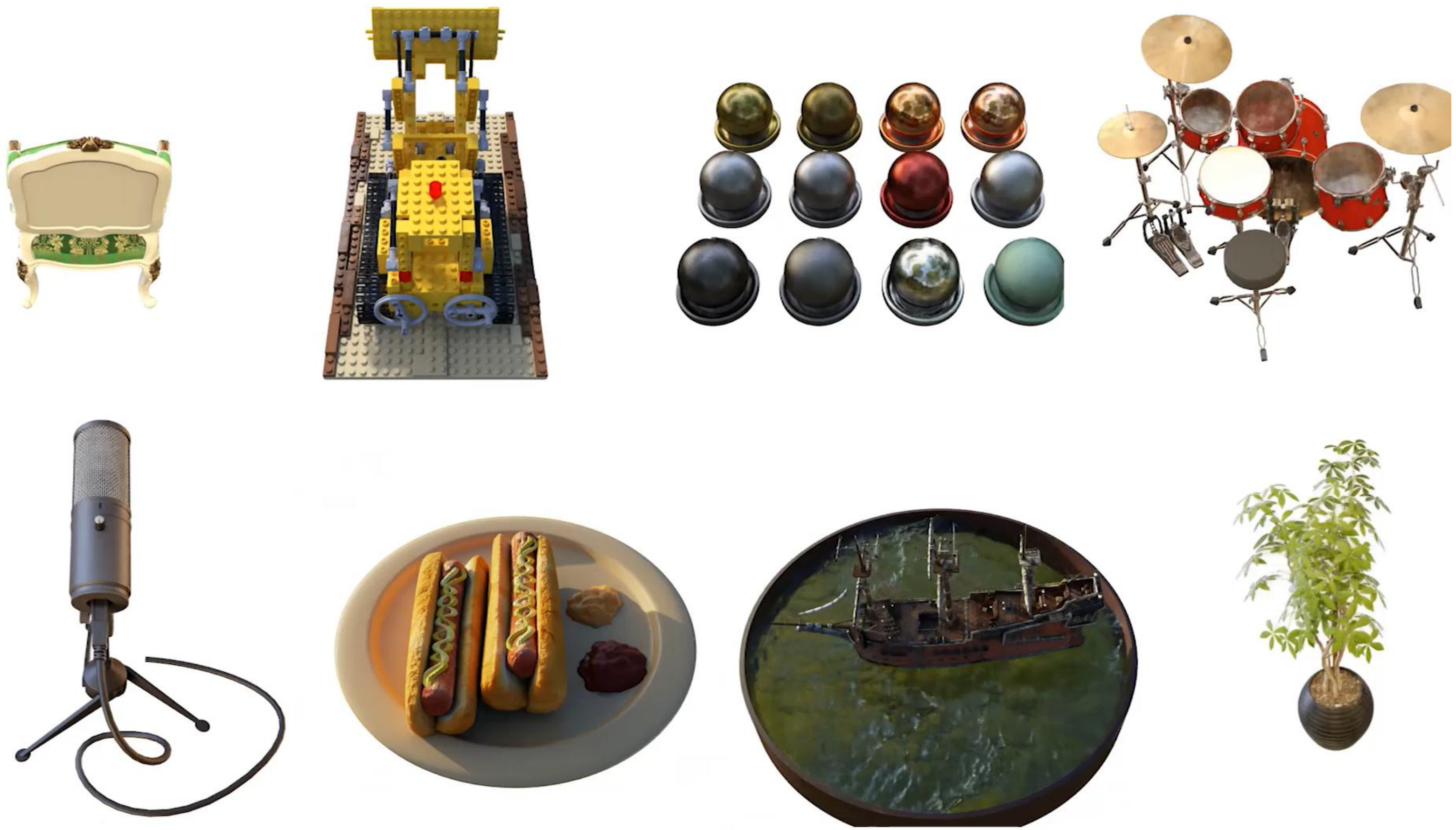


GT1
First image
Patchify
ViT
encoder
CNN
Shared
weights
Transformer
Decoder1
Head1
Pointmap
$X^{1,1} \in \mathbb{R}^{W\times H\times 3}$
Confidence
$C^1 \in \mathbb{R}^{W\times H}$
Common reference
frame of image I_1
Camera1
Cross-attention
GT2
Second image
Patchify
ViT
encoder
CNN
Transformer
Decoder2
Head2
Pointmap
$X^{2,1} \in \mathbb{R}^{W\times H\times 3}$
Confidence
$C^2 \in \mathbb{R}^{W\times H}$
Camera2
scale-invariant
regression L1 loss

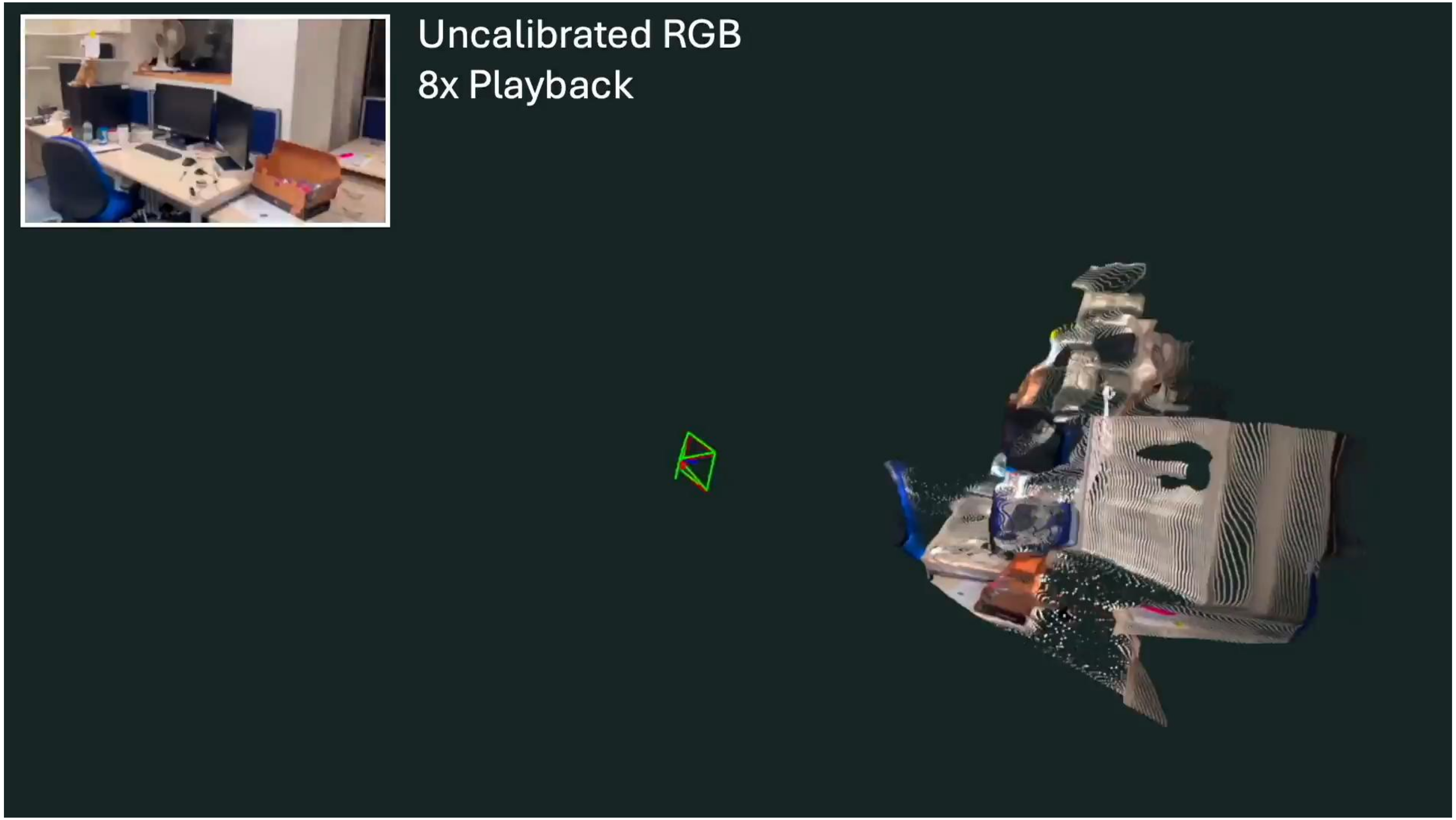
Neural radiance field (NeRF)



NERF is slow (0.1 FPS)

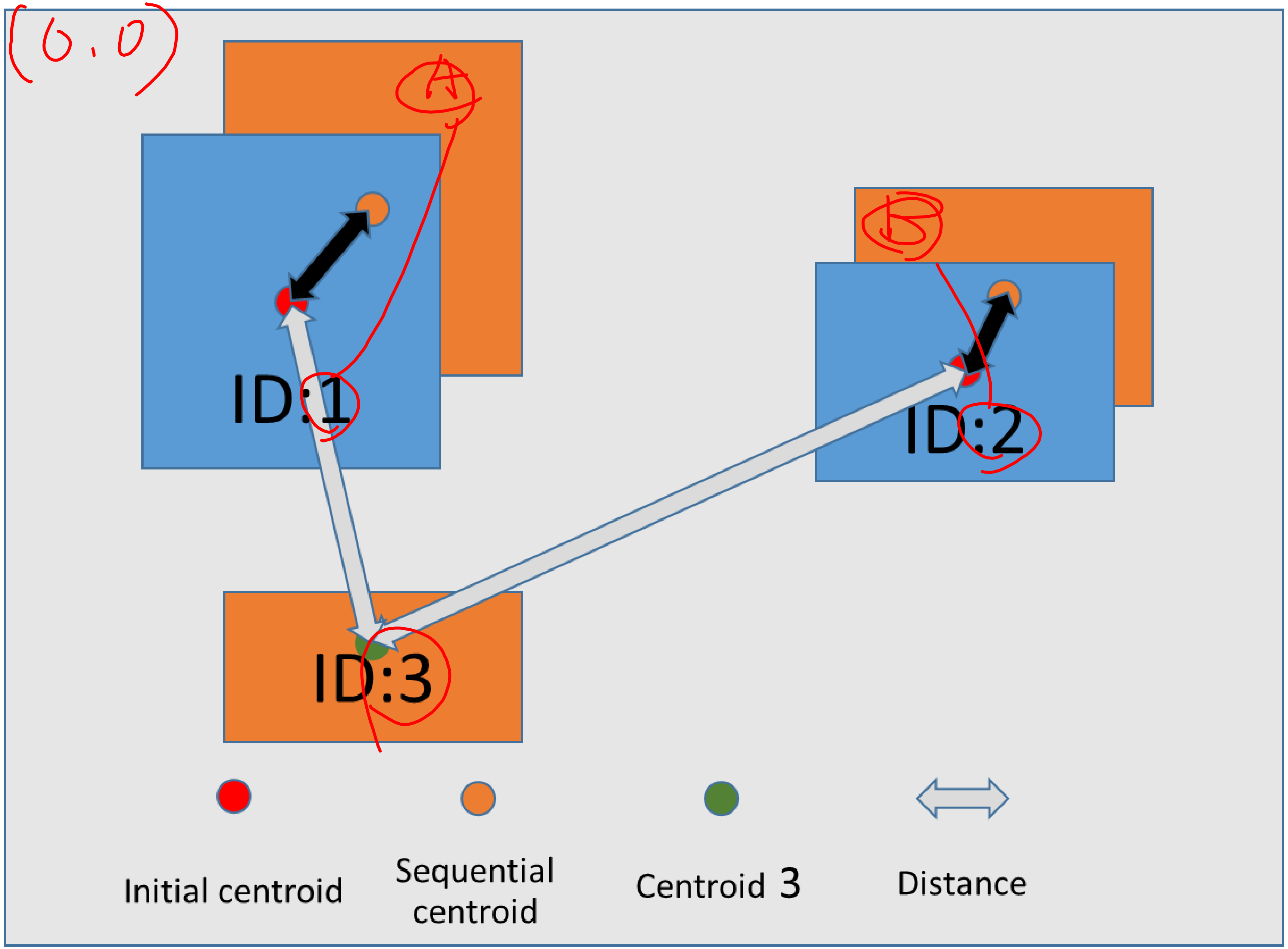


Mast3r-slam



Object tracking

1. Object detection (bounding box)
2. Object association (Hungarian association algorithm)



(0.0)
A
B
ID:1
ID:2
ID:3
Initial centroid
Sequential centroid
Centroid 3
Distance

Hungarian algorithm

1. Row Reduction by row minima
2. Column Reduction by column minima
3. Zero Coverage
4. Iterative Refinement (if needed)
5. Finding the Optimal Assignment

	1	2	…	j
ID: 1	C_{11}	C_{21}	…	C_{j1}
ID: 2	C_{12}	C_{22}	…	C_{j2}
…	…	…	…	…
ID: i	C_{1i}	C_{2i}	…	C_{ji}

Exercise

[Solve the assignment problem online - HungarianAlgorithm.com]

1+3+4
AD
BF

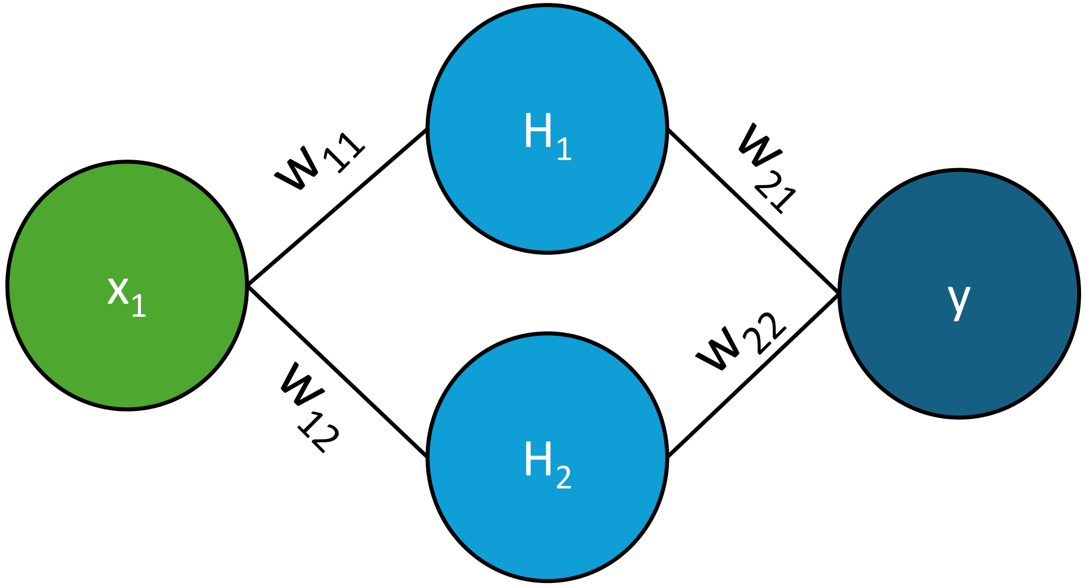
	Previous A	Previous B	Previous C
Next D	1 −1 = 0	3 −1 = 2 −1 = 1	1 −1 = 0
Next E	0	2 −1 = 1	4
Next F	6 −2 = 4	3 −2 = 1 −1 = 0	2 −2 = 0

4
1+3+0

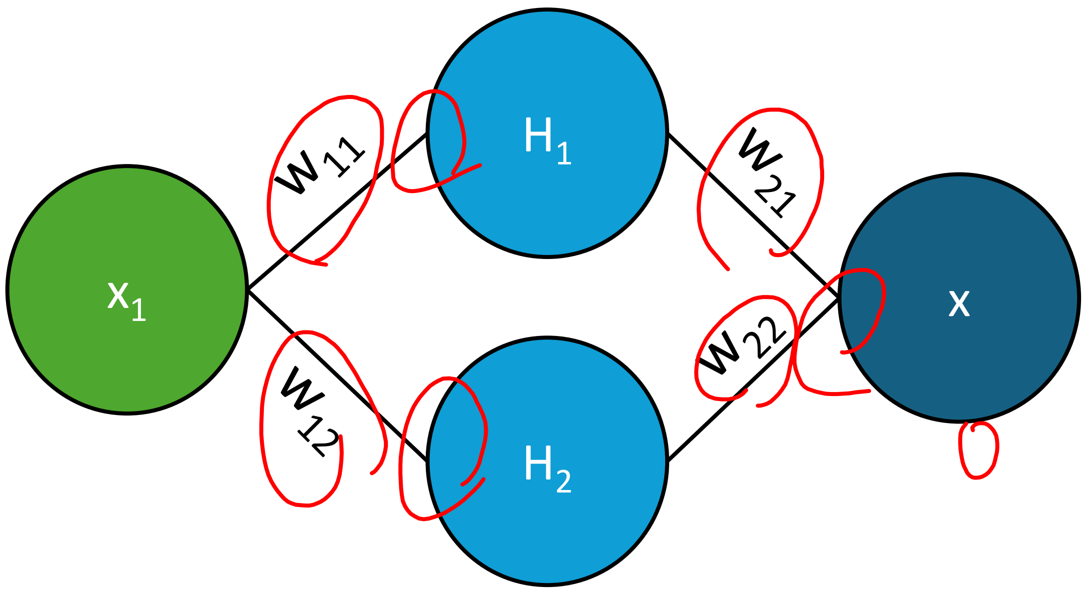
Lab 3a (until 2:45 pm)

- Understand the bounding box output
- ID association between two frames
- Visualize results

Auto-encoder-decoders



x	y
1	10
2	20
5	30

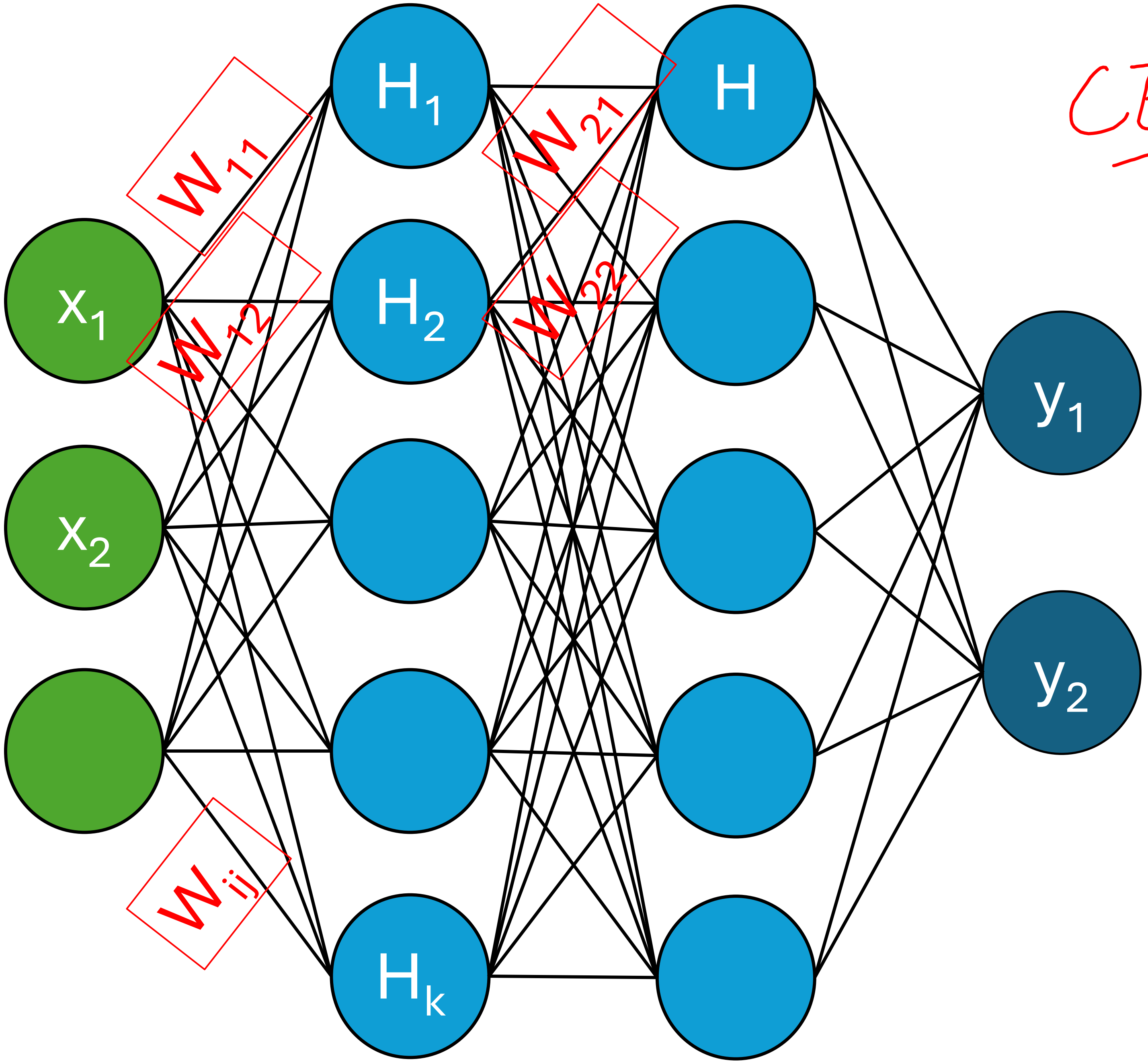


$$loss = \frac{(x_p - x)^2}{2}$$

$$\nabla = \frac{\partial loss}{\partial w}$$

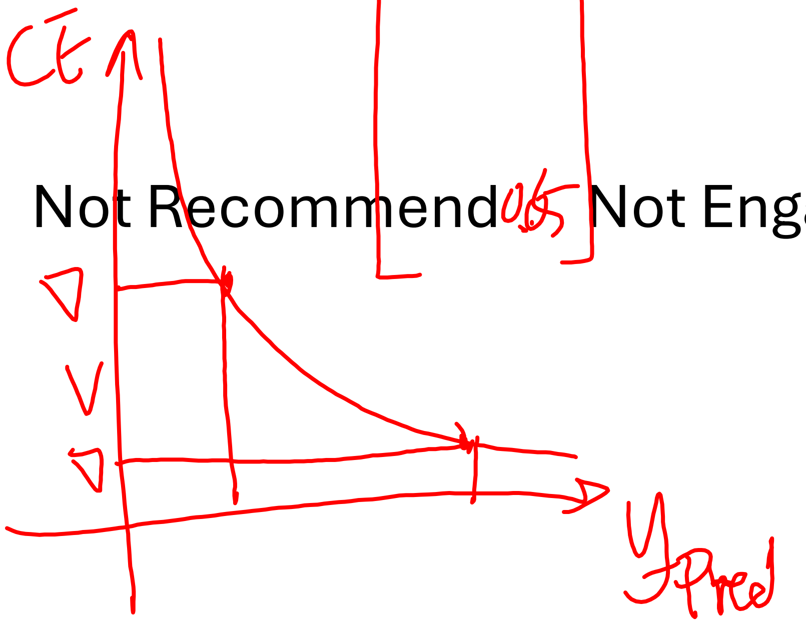
$$w' = w - \nabla \cdot LR$$

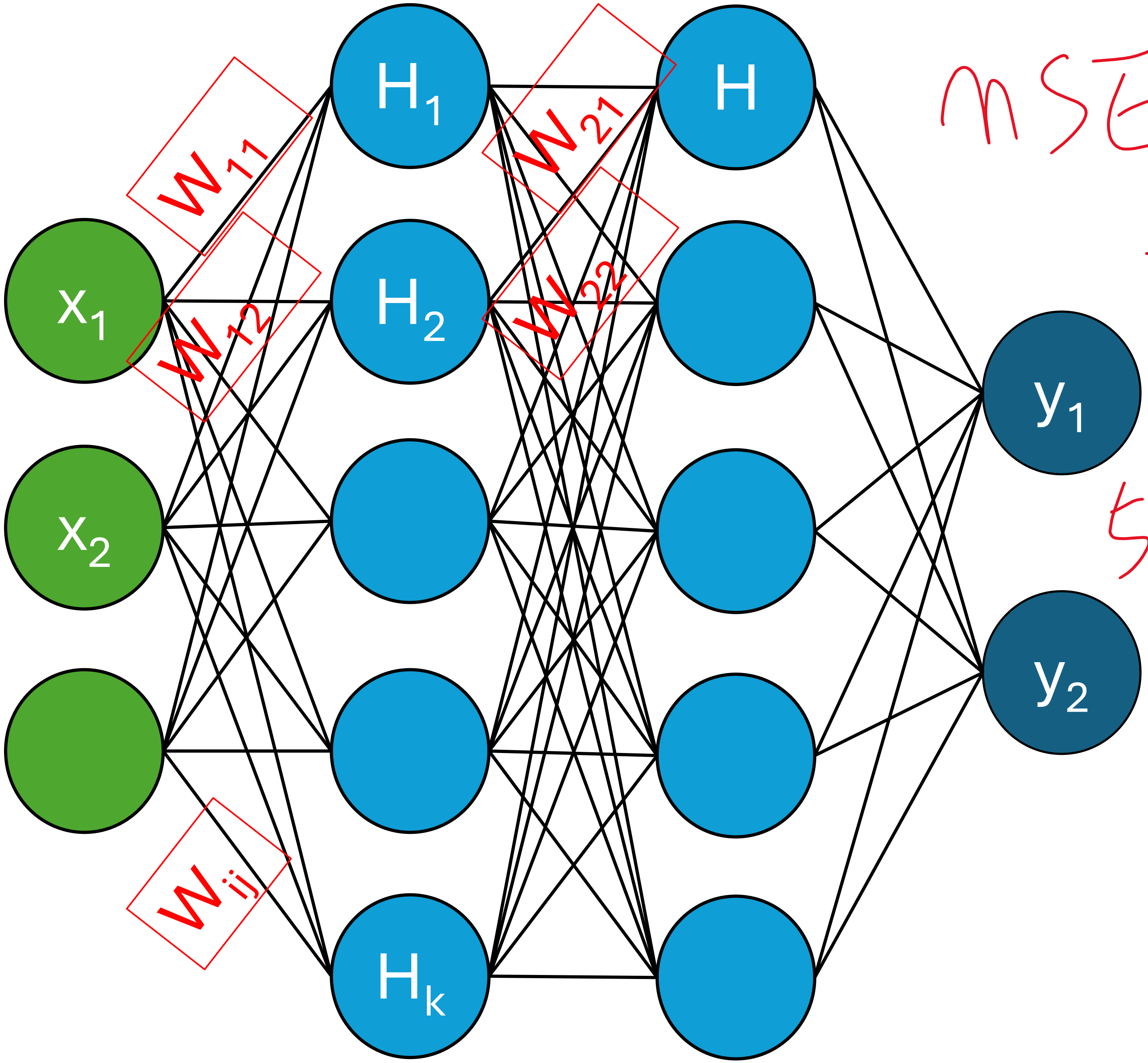
x	y
1	10
2	20
5	30



$$CE = -\log(y_p)$$

	y_{pred}		y_{true}
Recommend	0.95	Engaged	1
Not Recommend	0.05	Not Engaged	0





$$MSE \frac{(y_p - y_{gt})^2}{n}$$

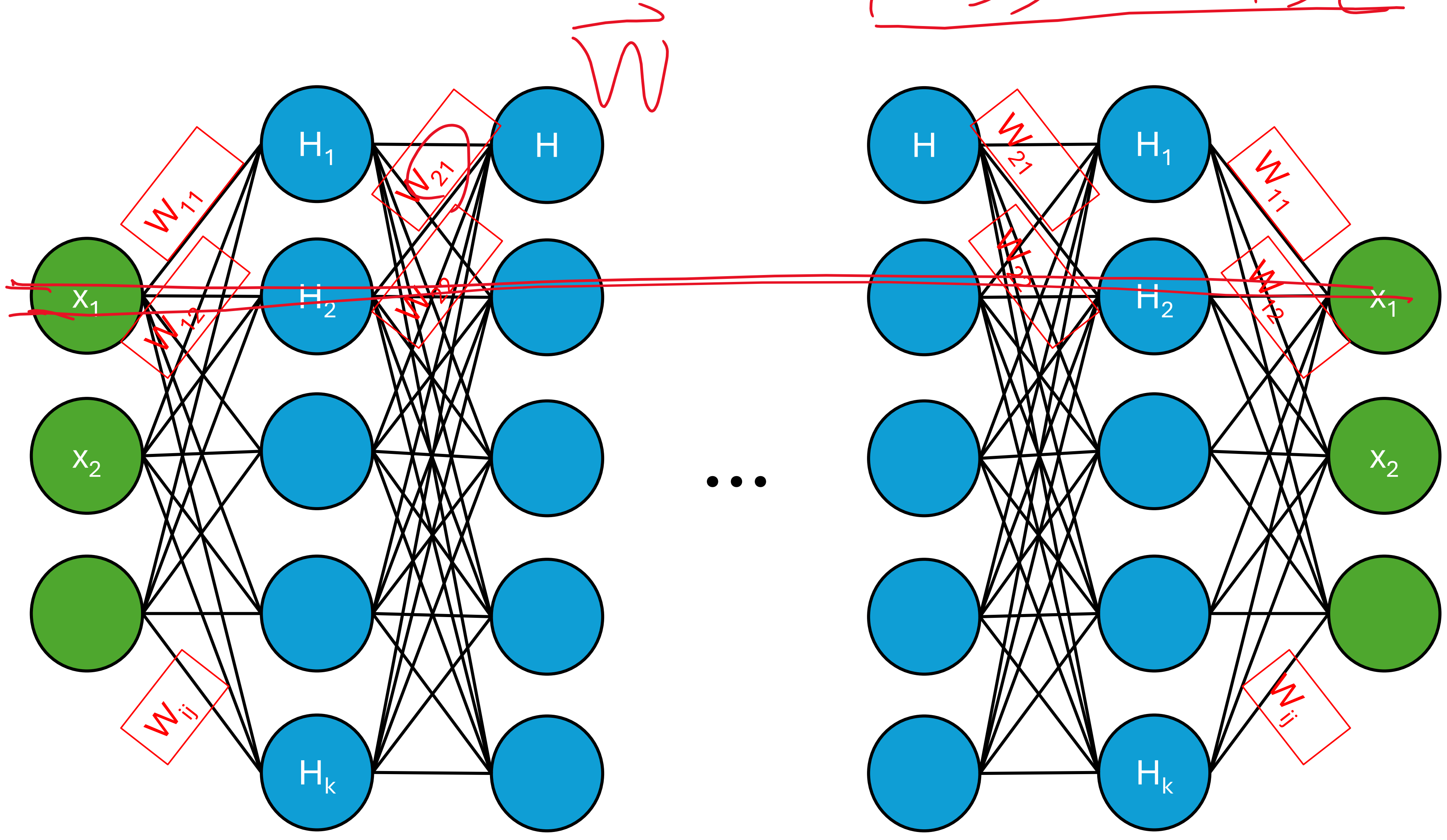
height

5'11''

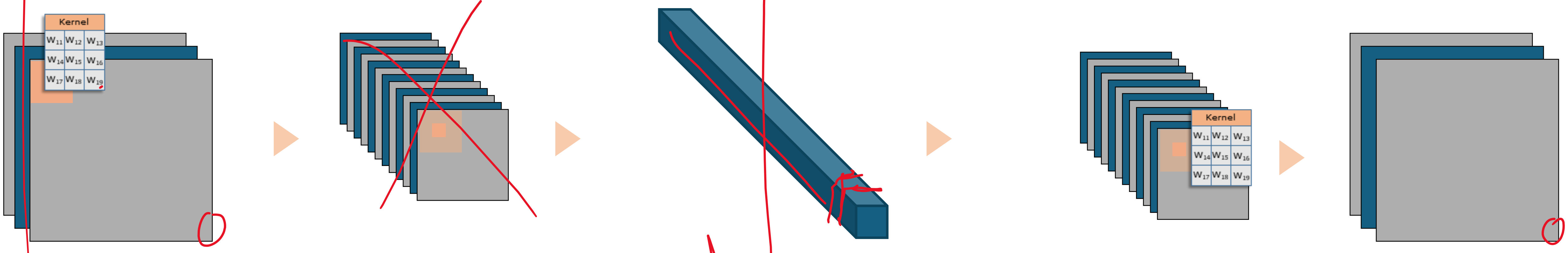
y_{gt} 6ft

Auto encoder decoder

Loss = MSE



CNN Auto-encoder-decoder



regression

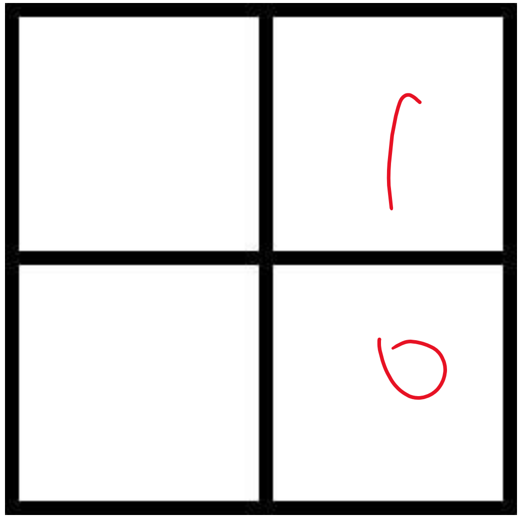
Input	Encoder	Bottleneck	Decoder	Output

mse
MAE

Generating image is hard

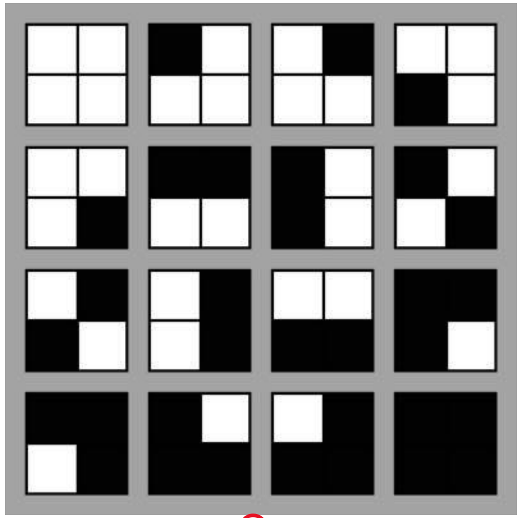
- Multi dimensional almost infinite

2^{2^2}



2^{8^2}

8



8

256^{1024^2}

0.2

28

512

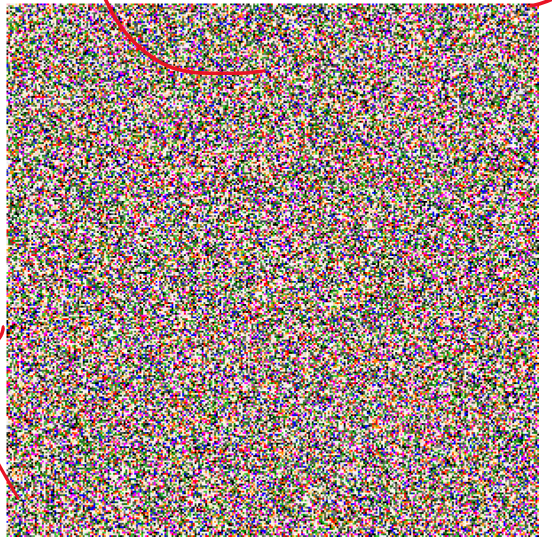
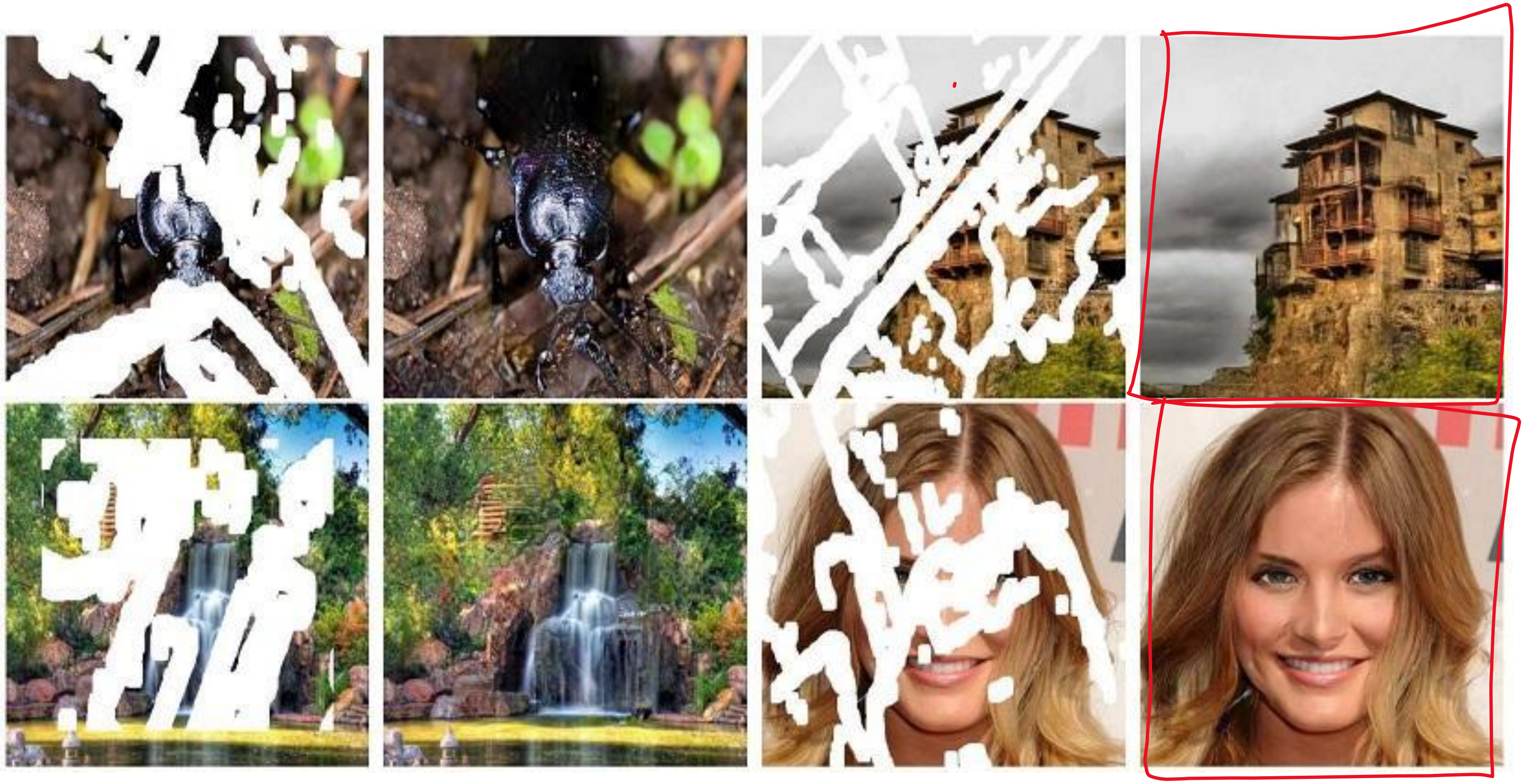


Image inpainting



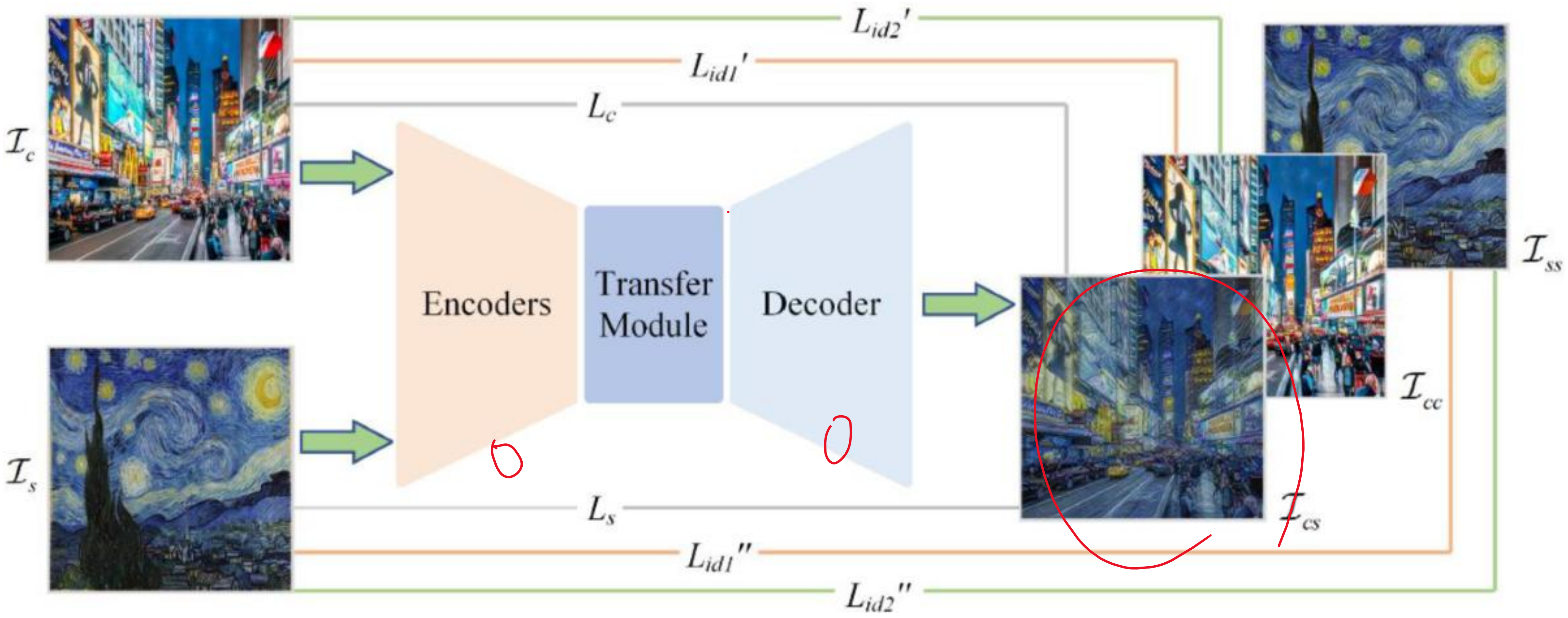
Old picture coloring

B → → C

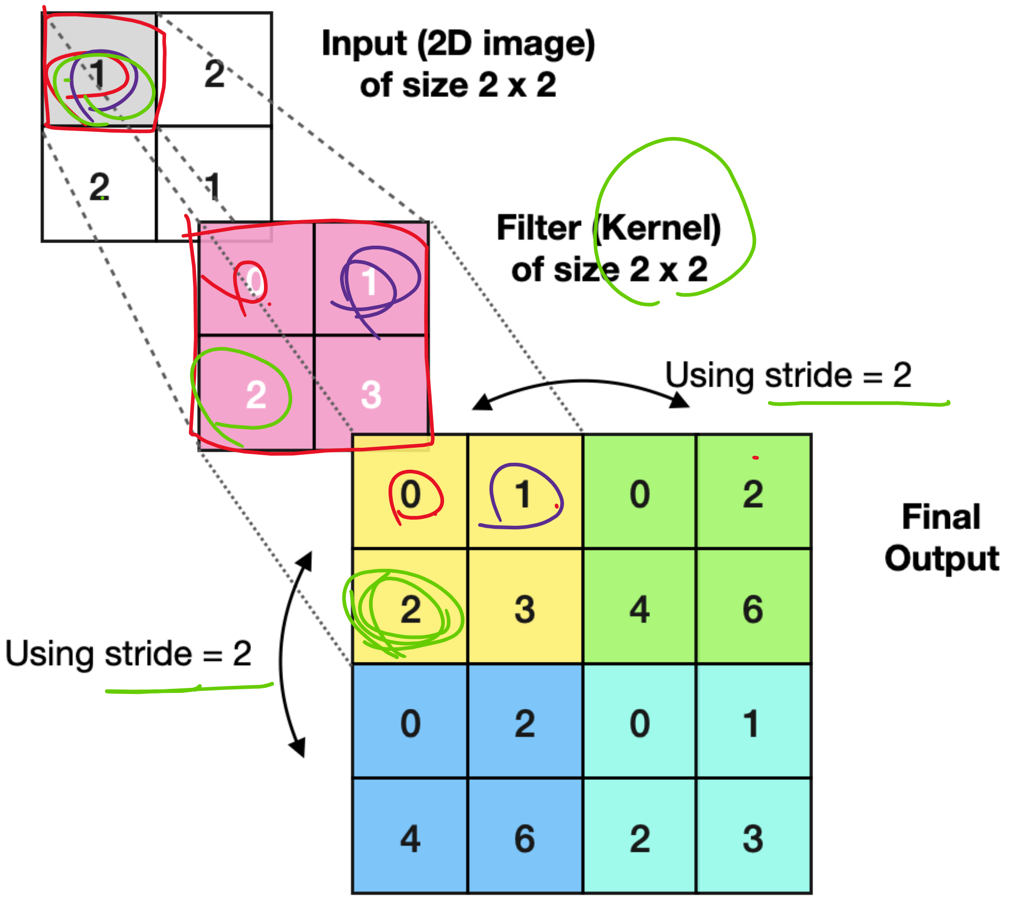
W

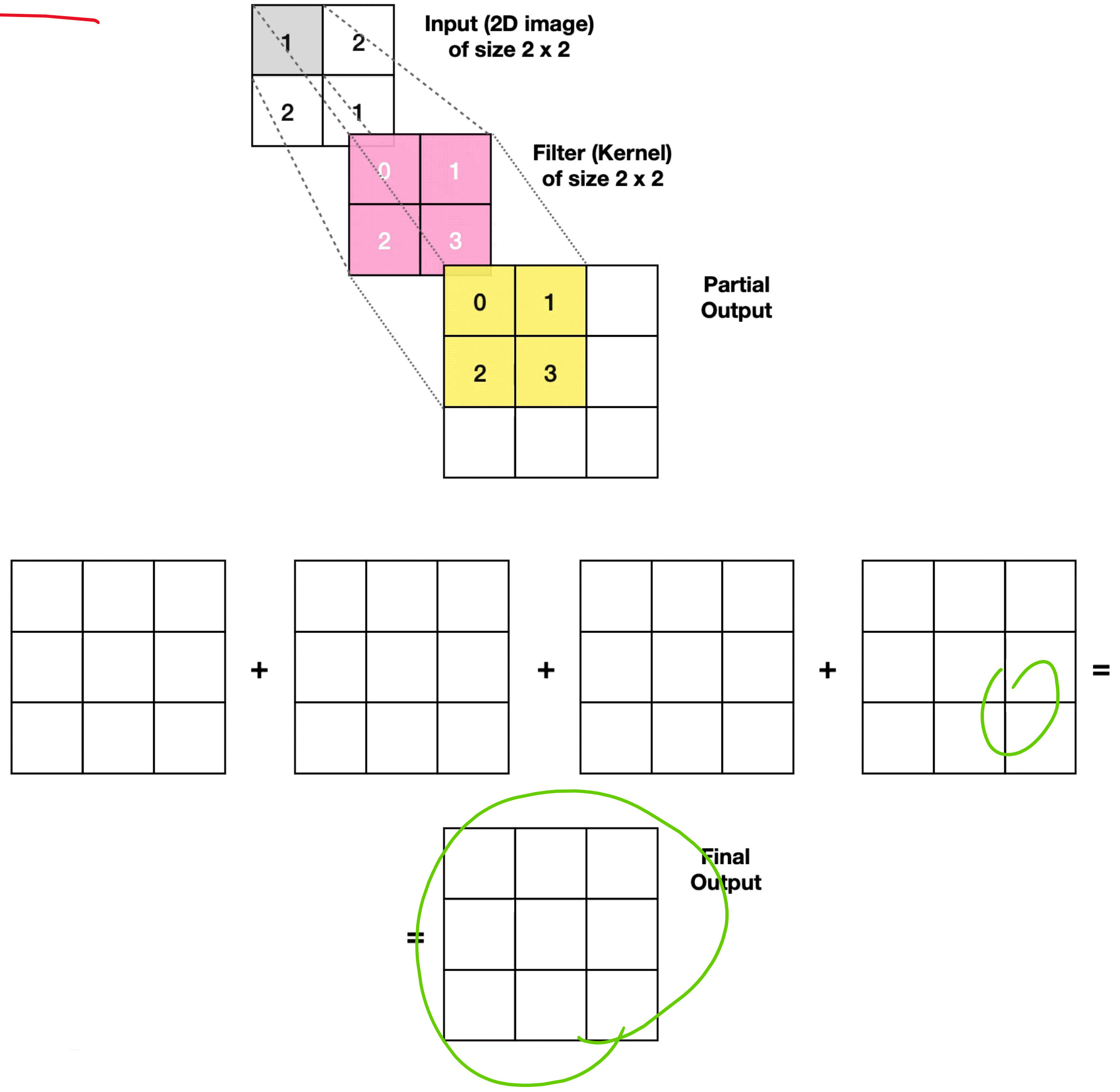
0~255

Style transfer



Transpose Convolution





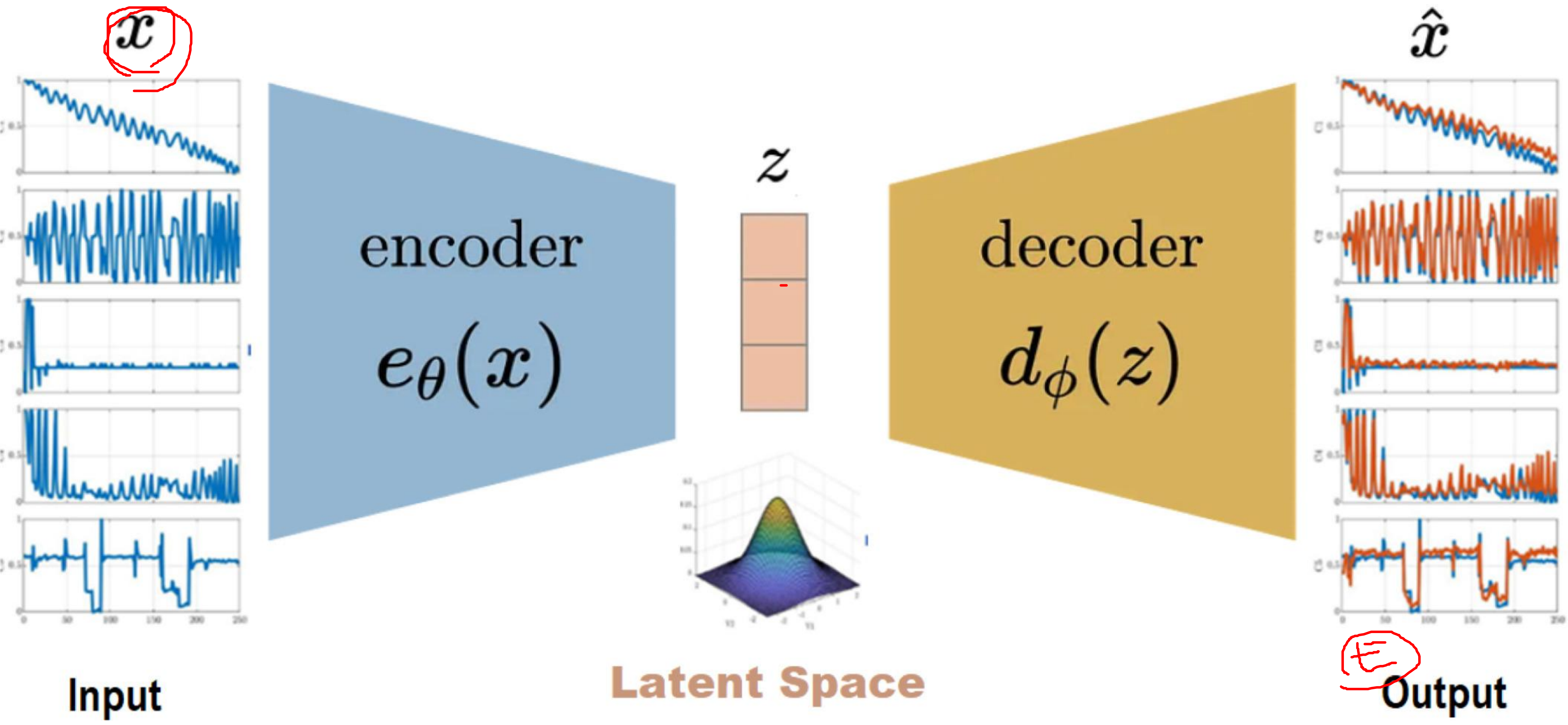
In front of a Chinese temple (1910-1919)

C

B

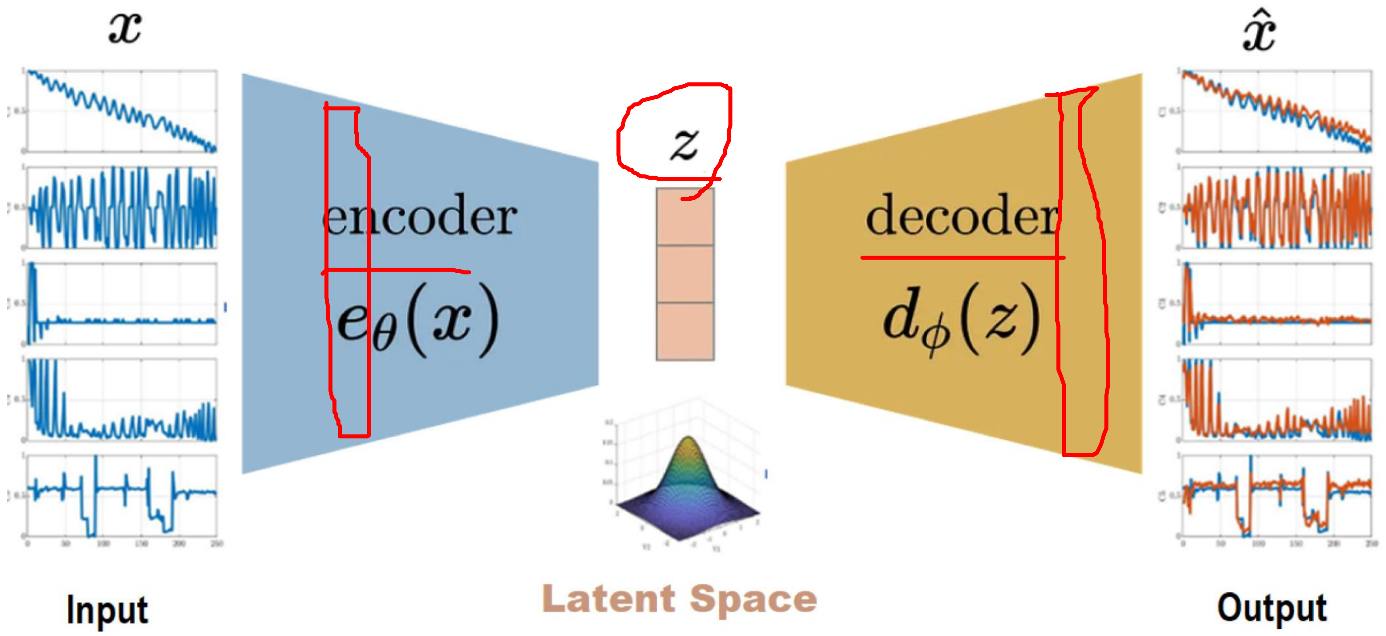
Video generator

- Seed dance
- Google veo
- OpenAI Sora
- META Emu


x
$\hat{x}$
encoder
$e_\theta(x)$
z
decoder
$d_\phi(z)$
Input
Latent Space
Output

Applications

- Feature extraction
- Data compression
- Image search
- Topic modeling
- Image coloring
- Translation
- ~~Anomaly detection~~



AI challenges

- Dataset
  - Supervised
  - Unsupervised
  - Self-supervised
- Model design
  - Data structure
  - Loss function
  - Neural network architecture

$NN(x) = y$

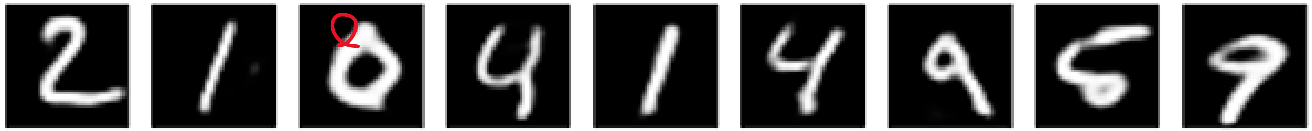
$Autoencode(x) = x$

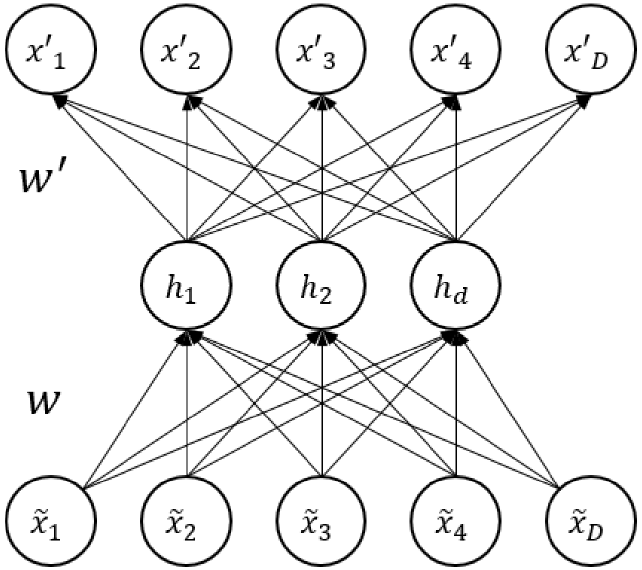
Lab 3b (until 4:00 pm)

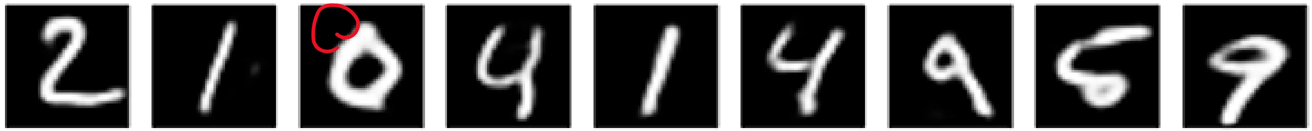
- Reconstruction
- Denoising
- Anomaly detection

Reconstruction

- Activation function
- Loss function
- Metric
- Optimizer
- Data set format

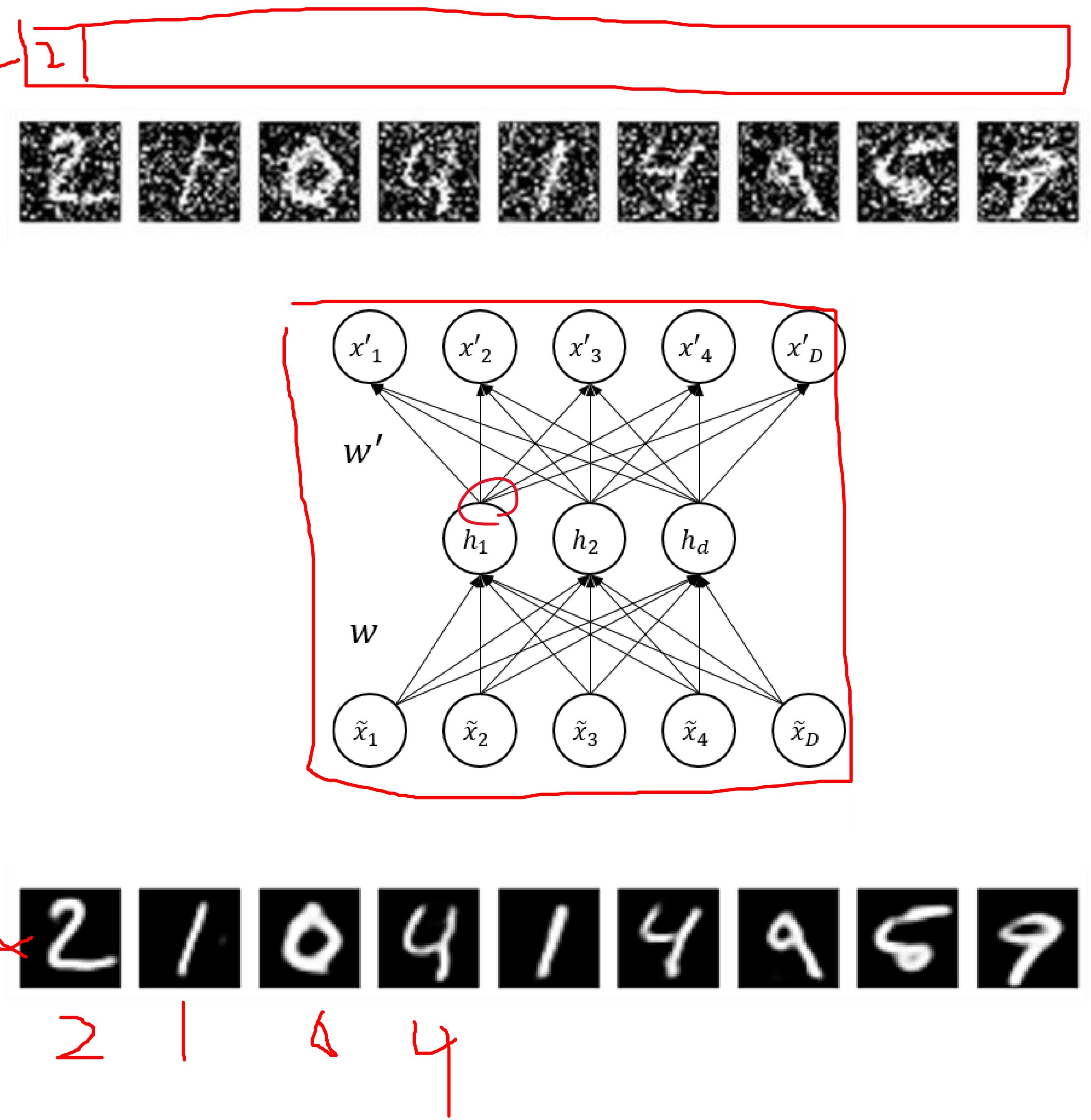






Denoising

- Activation function
- Loss function MSE
- Metric
- Optimizer
- Data set format

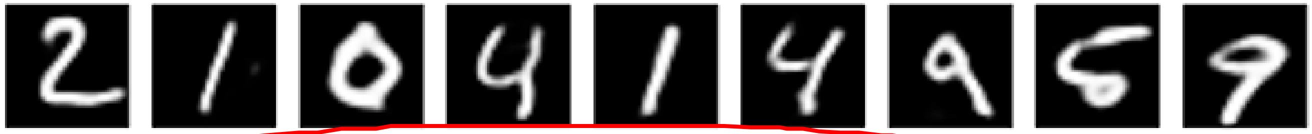


What if we keep adding noise

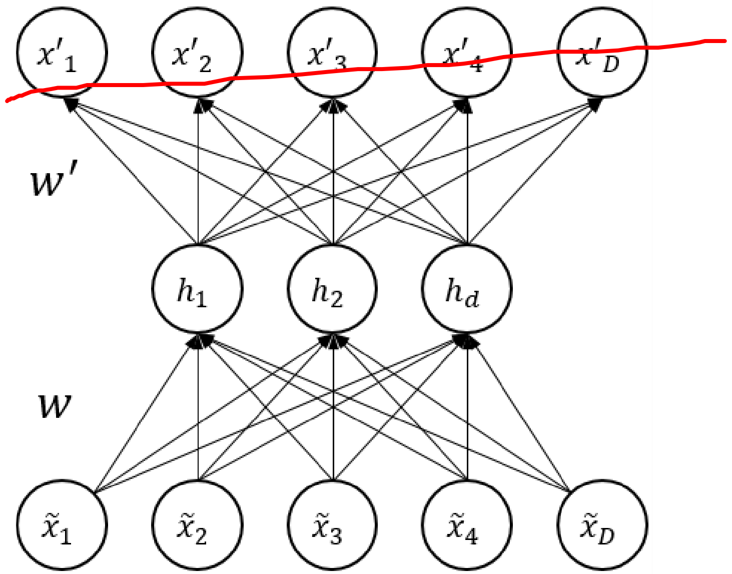
Generated Samples

Take random noise and generate samples

Anomaly detection

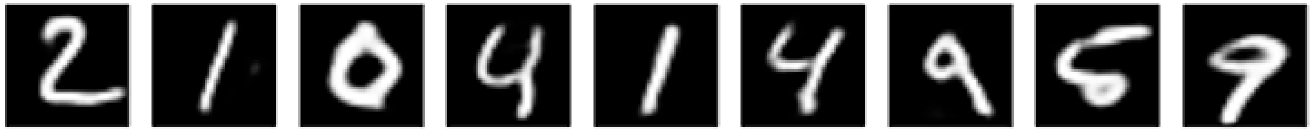


$\frac{1}{D}$



- Activation function
- Loss function
- Metric
  - MAE
  - $\frac{TP}{n}$
- Optimizer
- Data set format
  - Only train on normal data
  - Abnormal labels

θ



E > threhold

std + meam

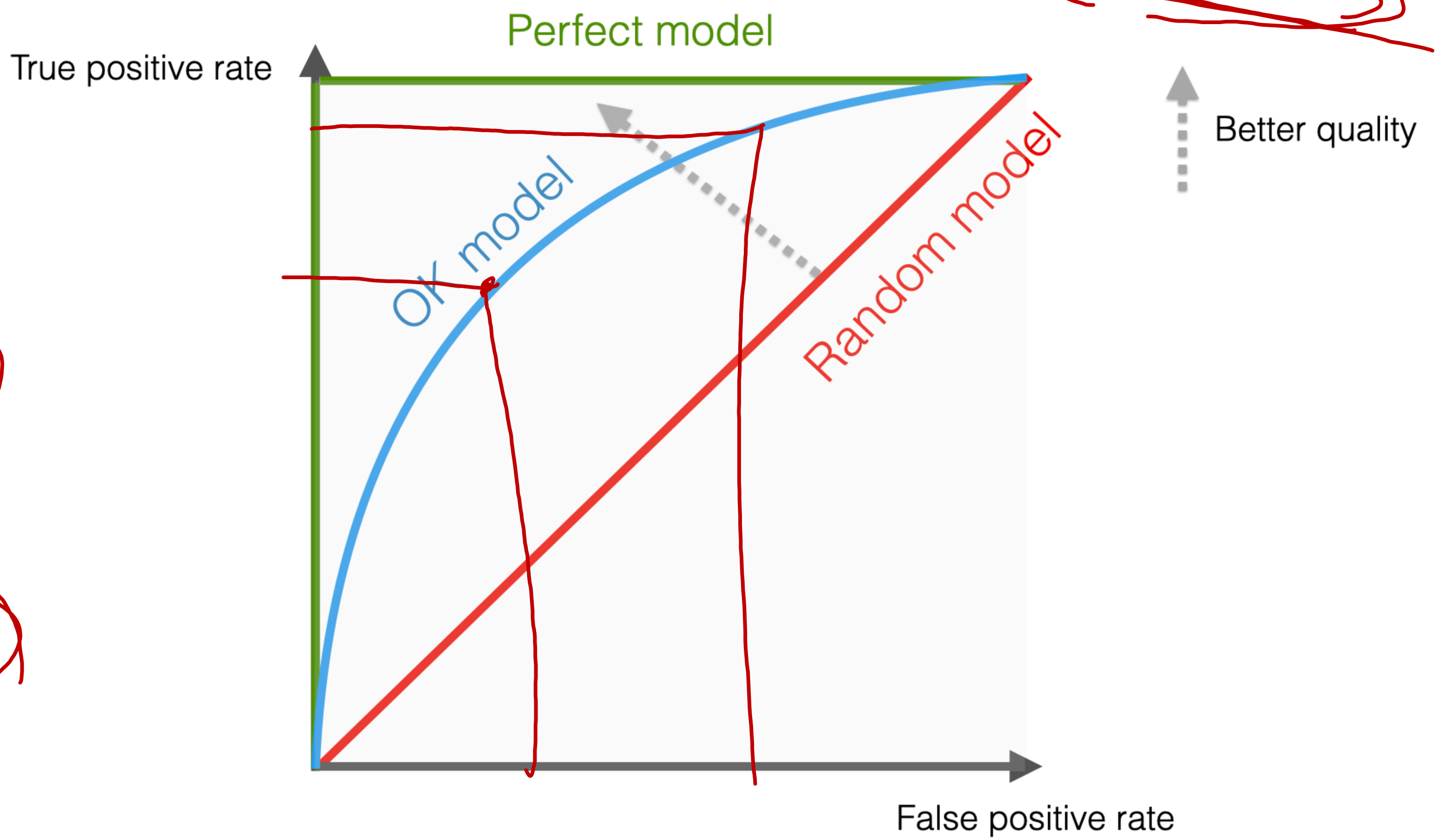
Receiver operating characteristic (ROC) area under the curve (AUC)

y_p 0.56 c

0.5 0.4 0.3

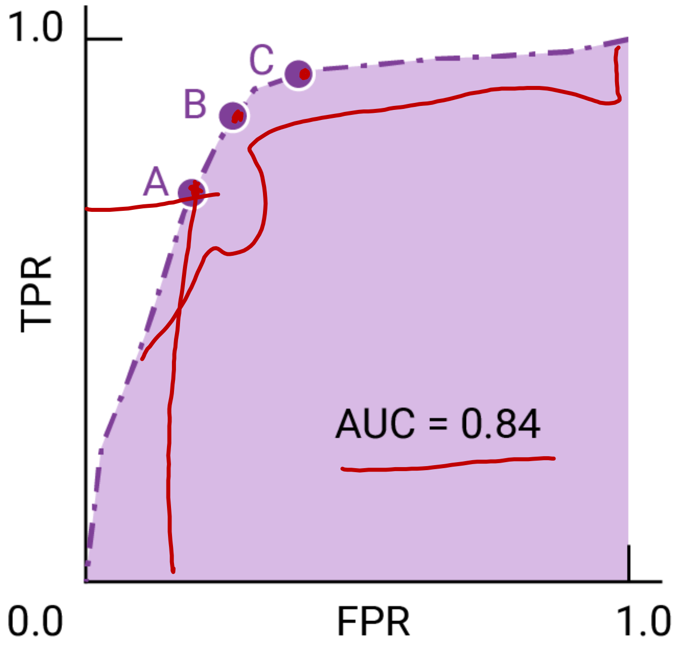
$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$



Binary classification

- High cost of false positives: low FPR (Point A), accepting a reduced TPR to minimize false alarms.
- High cost of false negatives: Choose an aggressive threshold with high TPR (Point C), accepting more false positives to avoid missed detections.
- Similar costs: Choose a balanced threshold (Point B) that trades off TPR and FPR more evenly.

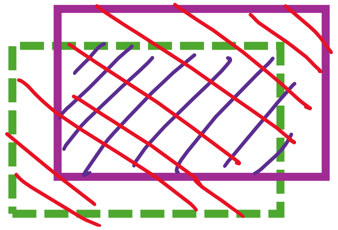


Object detection metric

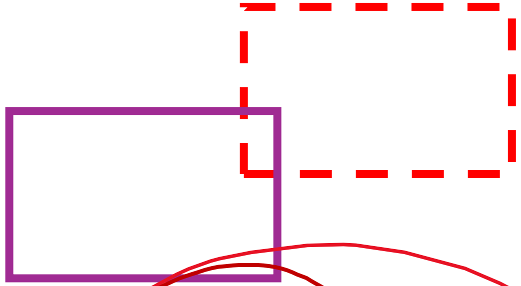
$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$

Ground Truth

Prediction



True **P**ositive (**TP**): IoU ≥ 50%



False **P**ositive (**FP**): IoU < 50%

No True Negative (TN)

False **N**egative (**FN**): IoU = 0

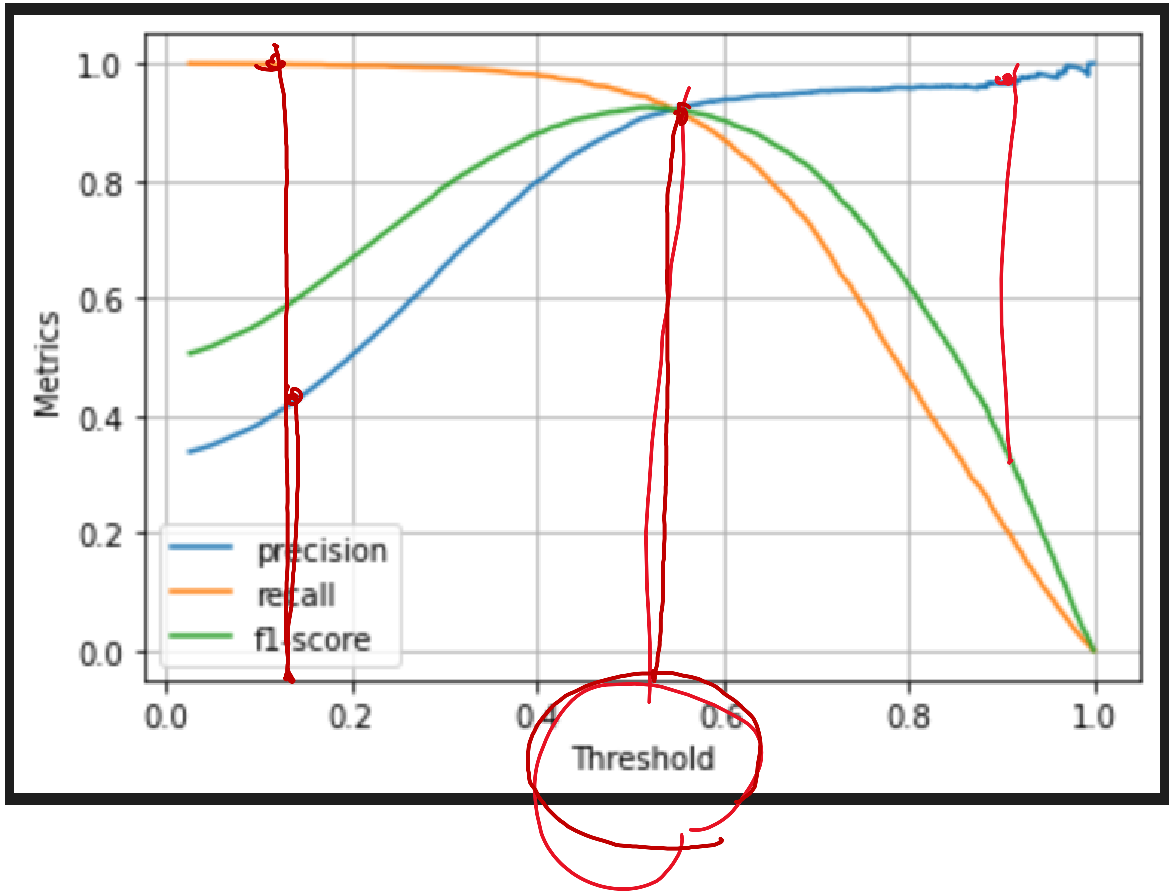
Prediction\Ground Truth	Positive	Negative
Positive	TP	FP
Negative	FN	TN

$$accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

$$precision = \frac{TP}{TP + FP}$$

$$recall = \frac{TP}{TP + FN}$$

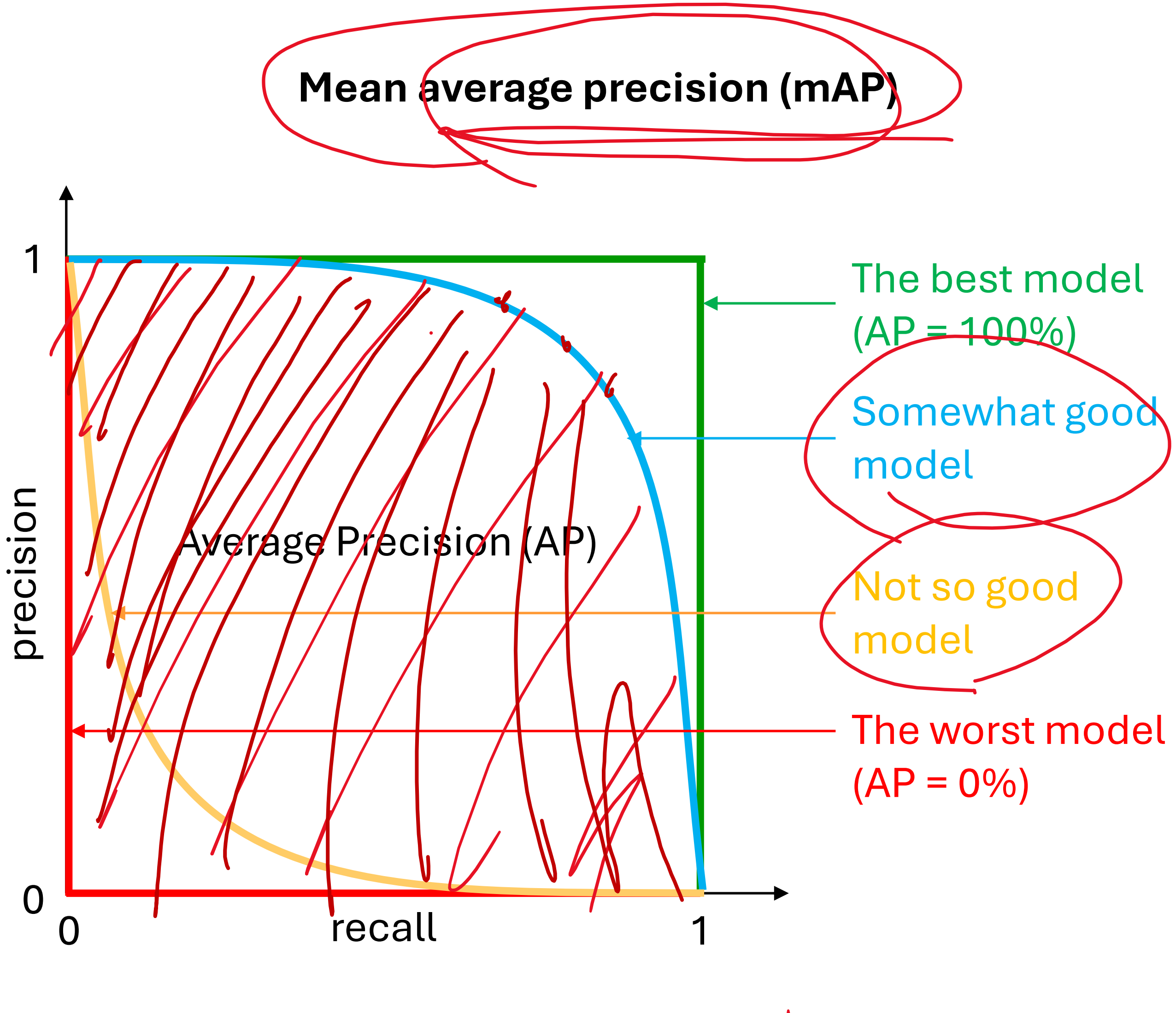
$$F1 = \frac{precision * recall}{precision + recall}$$



mAP

$$precision = \frac{TP}{TP + FP} = \frac{TP}{Predictions}$$

$$recall = \frac{TP}{TP + FN} = \frac{TP}{Ground\ Truth}$$

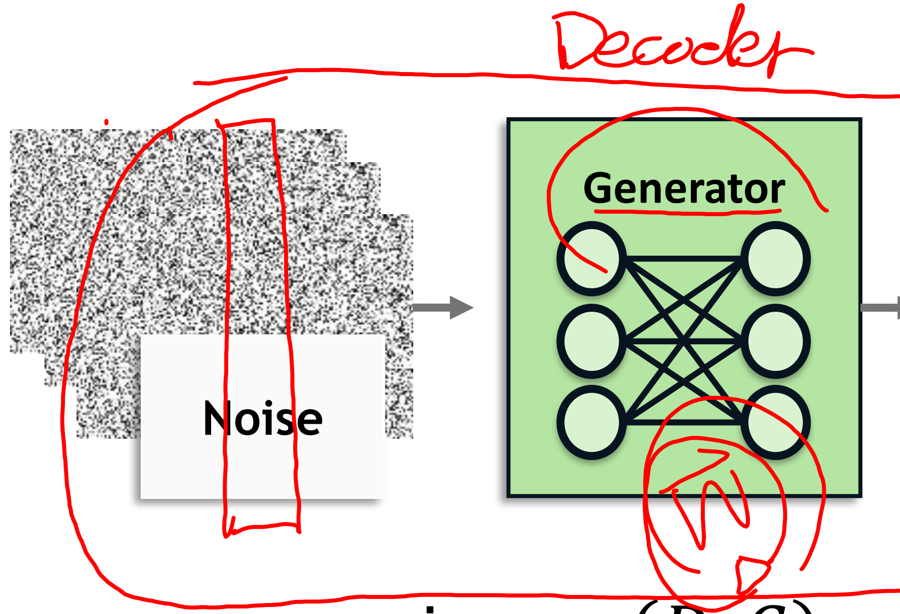


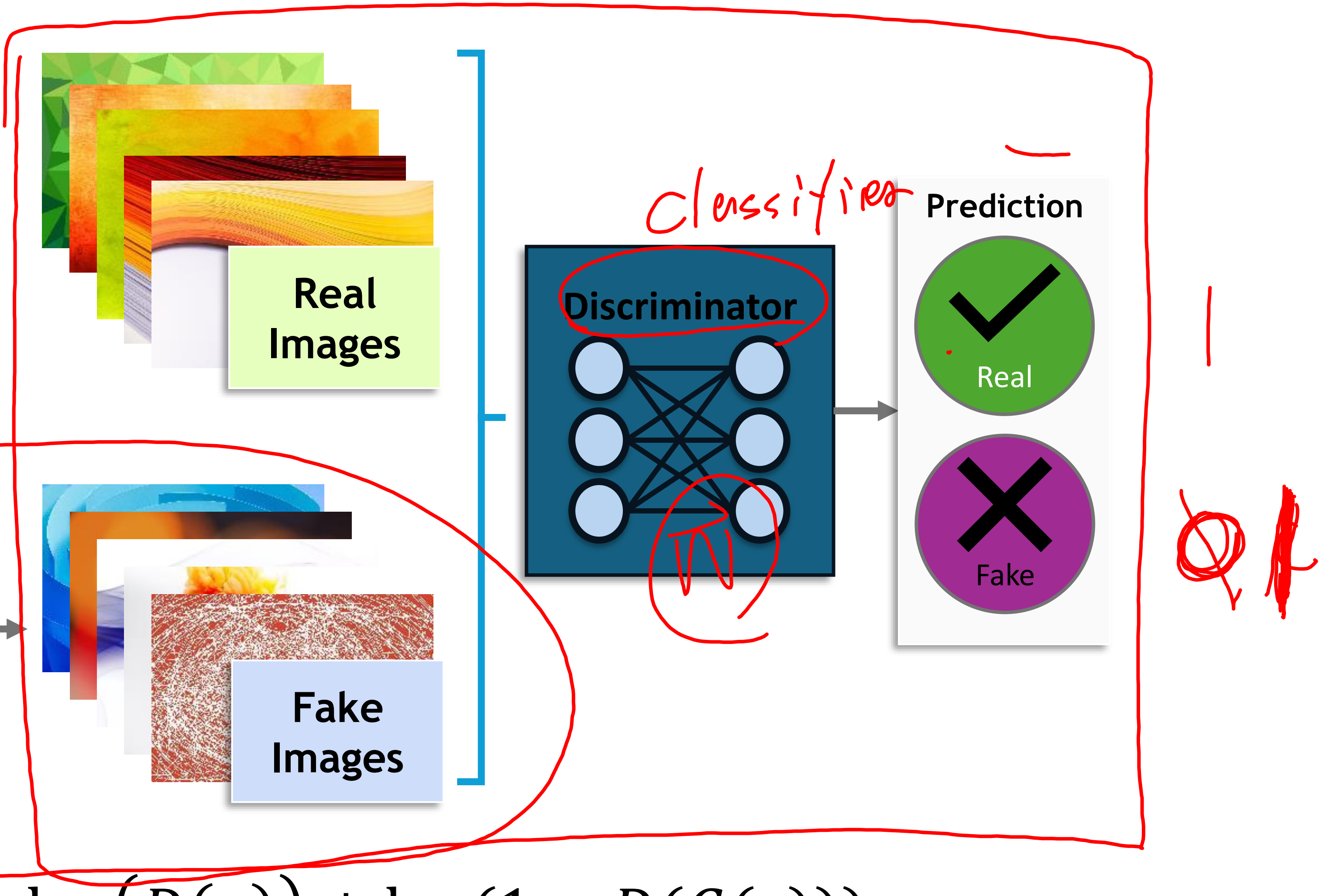
Lab 3c (until 4:40 pm)

- Anomaly detection
- No labels for threshold

Generative Adversarial Networks (GANs)

Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., ... & Bengio, Y. (2020). Generative adversarial networks. *Communications of the ACM*, *63*(11), 139-144.



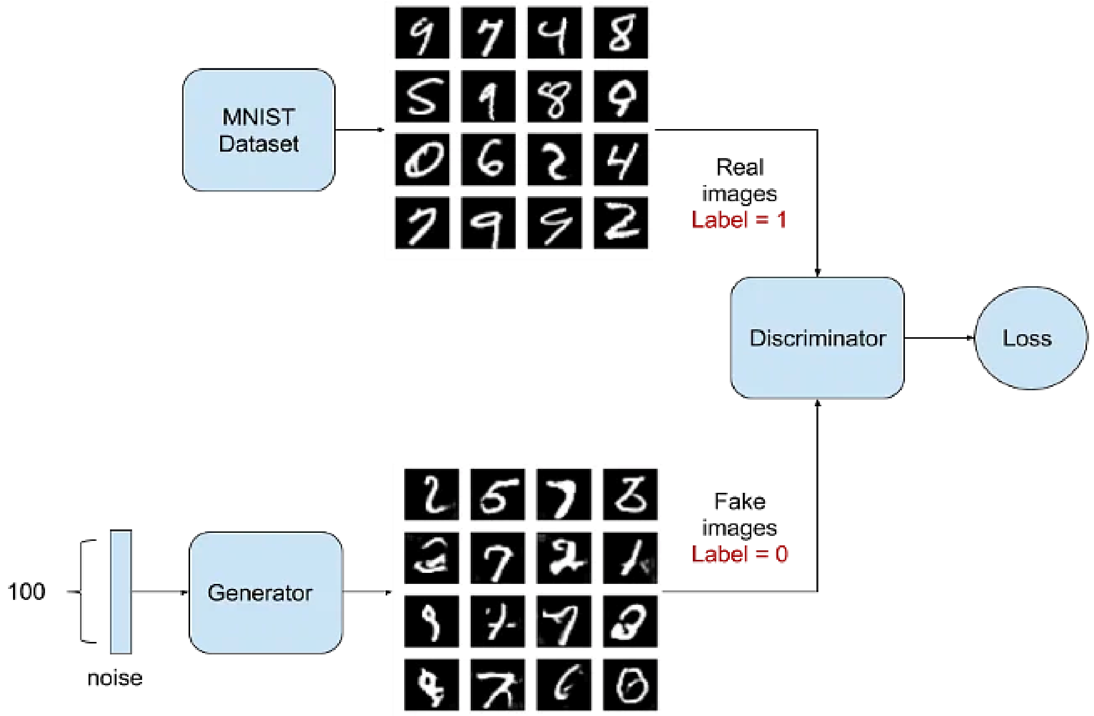


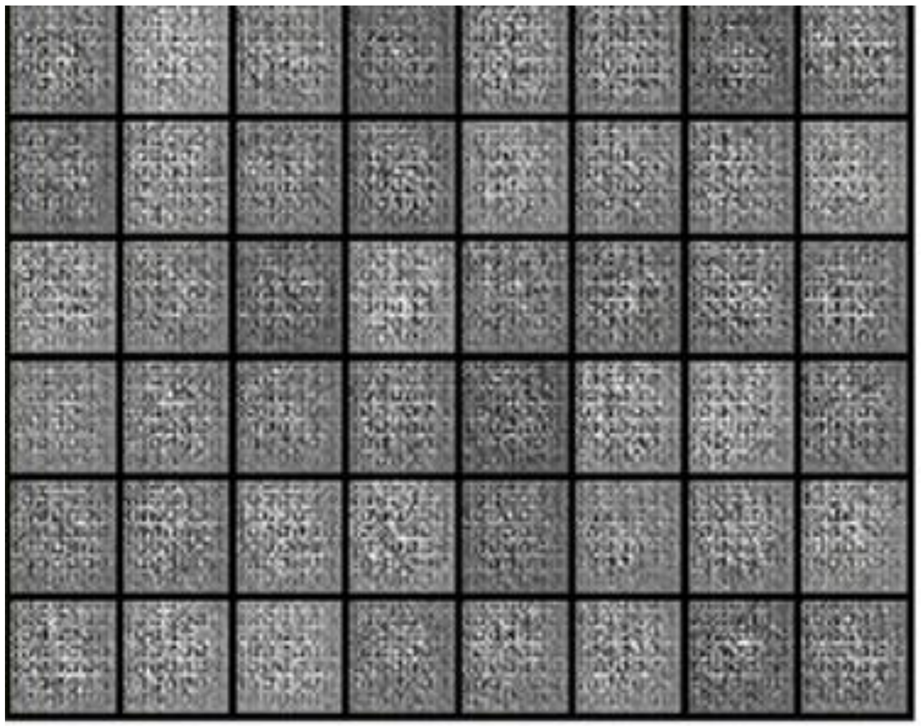
$$\min_{G}\max_{D}(D,G) = \log\big(D(x)\big) + \log(1 - D(G(z)))$$

Generative Adversarial Networks – Creating Images

GANs can generate essentially any form of data.

For the MNIST problem, GANs can produce a flat array of 0s and 1s.





Progression of generating MNIST examples as a GAN is trained

Generative Adversarial Networks – Deep Convolution GANs

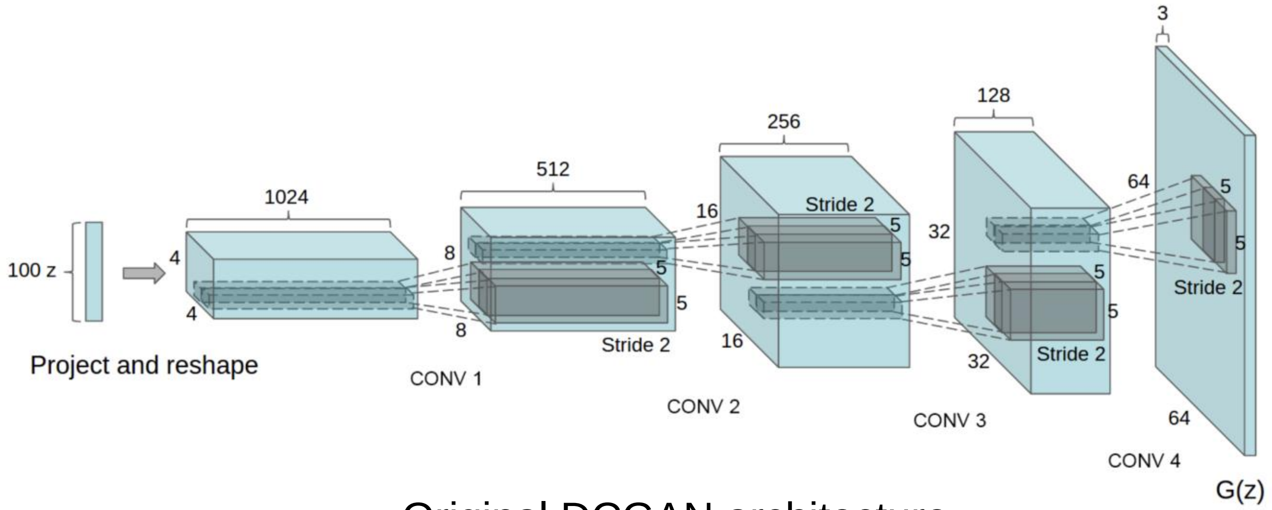
Recall that we have seen the transpose convolution. We can create a new type of generator, one that includes these transpose convolution.

DCGAN (**D**eep **C**onvolution **GAN**)

Input a random column vector

Passed through several transpose convolutions

Final result is a generated image!



Original DCGAN architecture

DCGAN – Exploring the latent space of images

With a trained DCGAN, we can manipulate the latent vector, z, to explore the space of images we can generate.

Latent Vector Space (adding the z vectors)

Pixel Space (adding the pixel values)

GAN APPLICATIONS

Figure 5: 1024 × 1024 images generated using the CELEBA-HQ dataset. See Appendix F for a larger set of results, and the accompanying video for latent space interpolations.

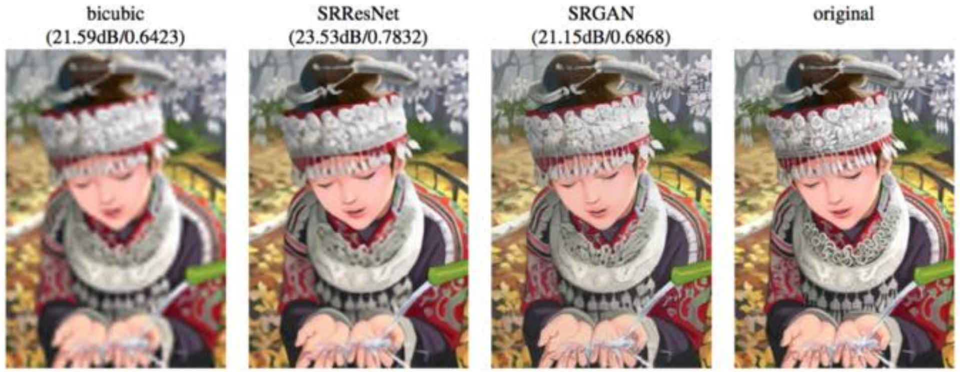
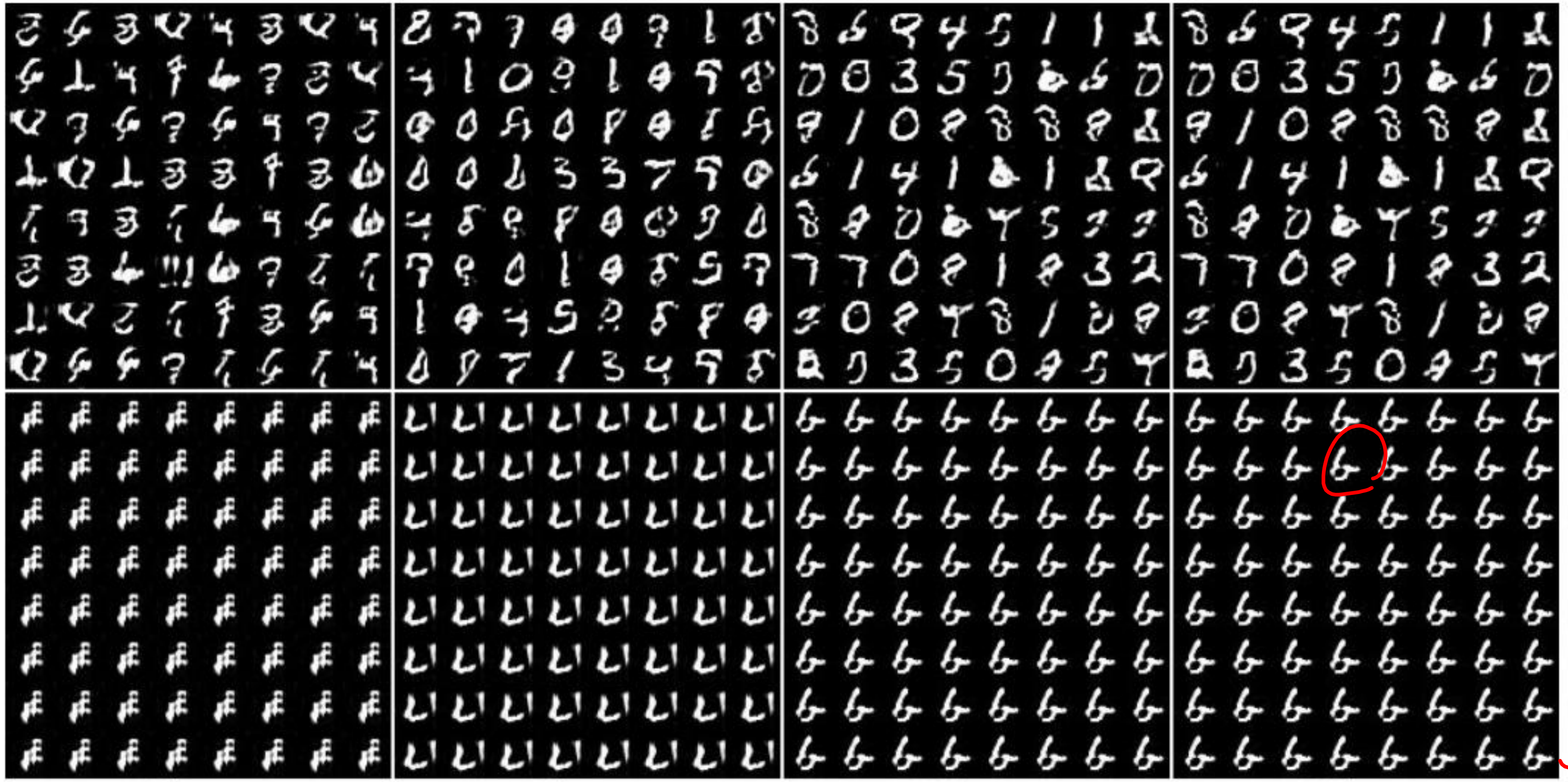


Figure 2: From left to right: bicubic interpolation, deep residual network optimized for MSE, deep residual generative adversarial network optimized for a loss more sensitive to human perception, original HR image. Corresponding PSNR and SSIM are shown in brackets. [4× upscaling]

pix2pixHD

GAN collapse

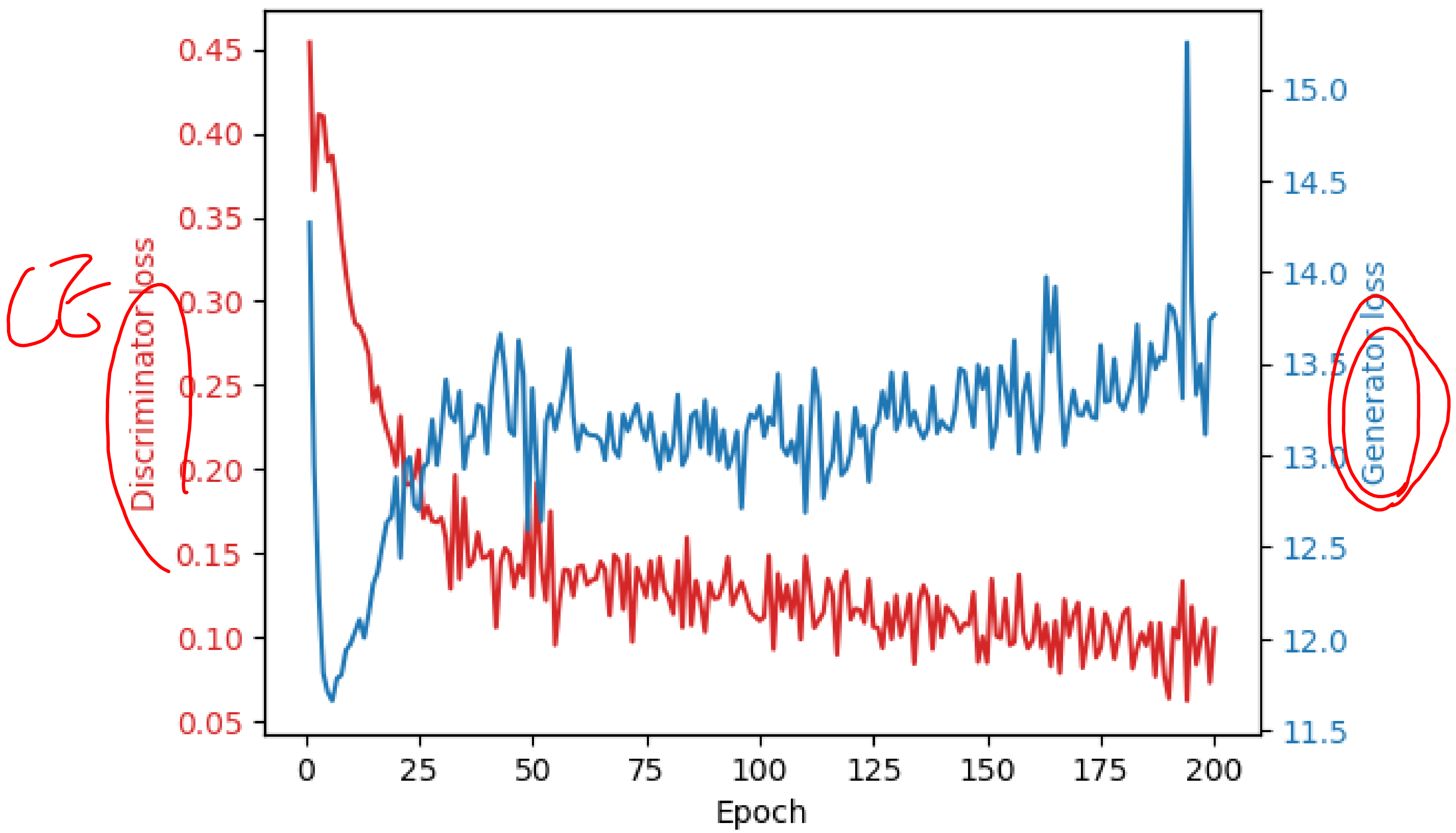


10k steps 20k steps 50K steps 100k steps

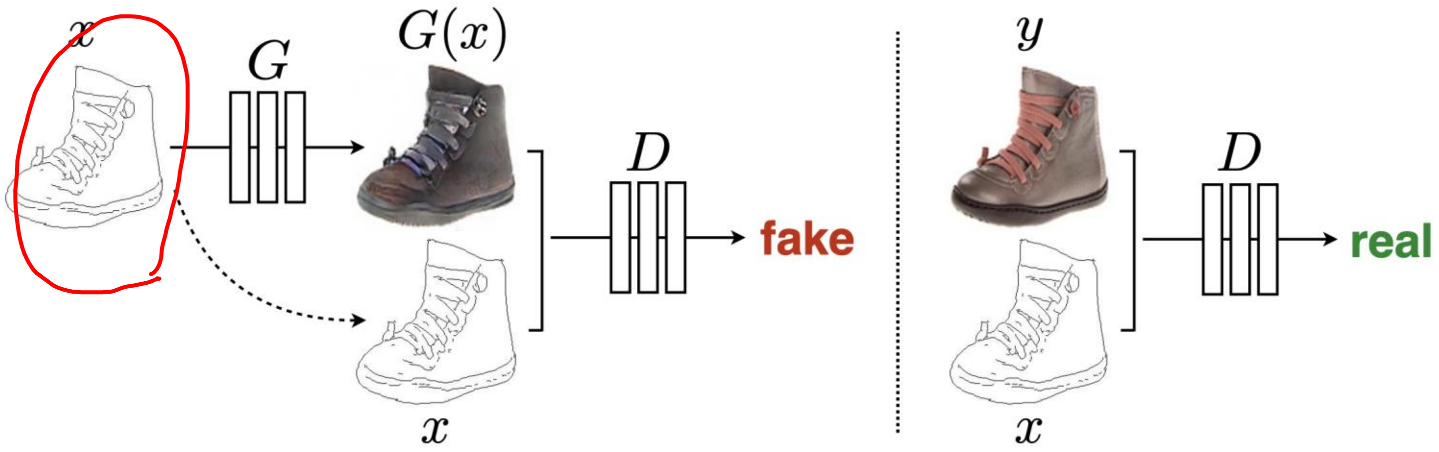
$G(\cdot)$

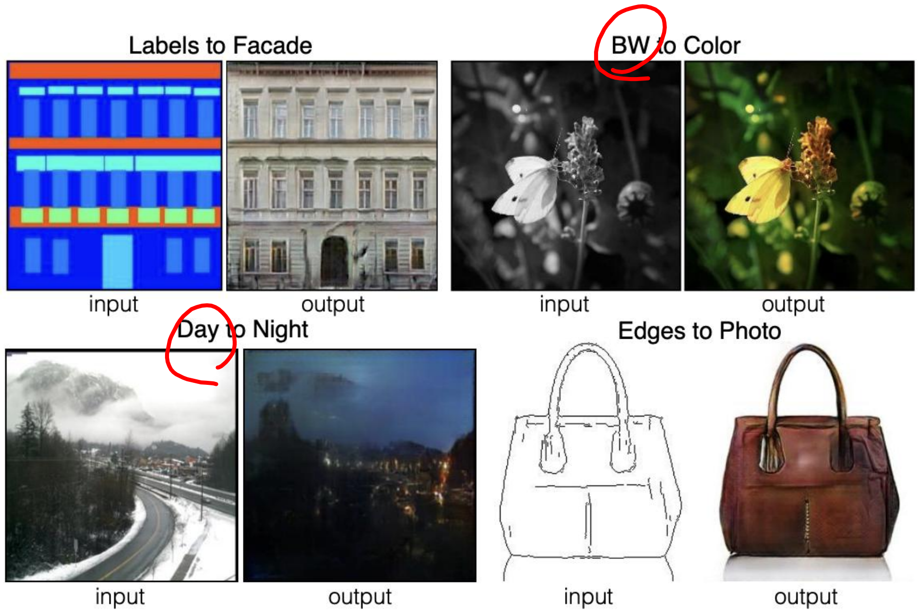
https://link.springer.com/article/10.1007/s12065-021-00665-z
https://arxiv.org/pdf/1611.02163

Insatiability



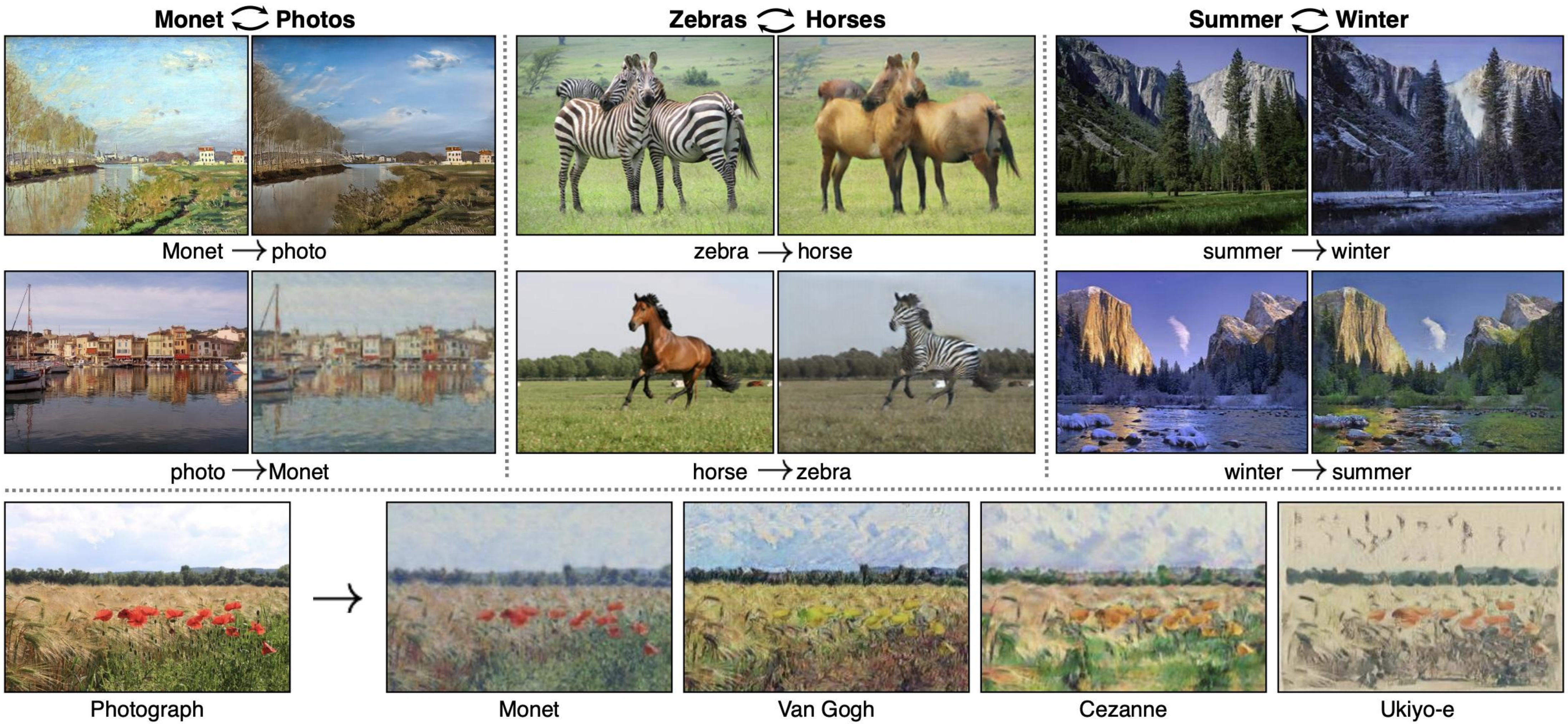
Conditional GAN (cGAN): Targeted Image generation





https://arxiv.org/abs/1611.07004, https://arxiv.org/abs/1610.09585

CycleGAN



Monet ⇄ Photos
Zebras ⇄ Horses
Summer ⇄ Winter
Monet → photo
zebra → horse
summer → winter
photo → Monet
horse → zebra
winter → summer
Photograph
Monet
Van Gogh
Cezanne
Ukiyo-e

CycleGAN

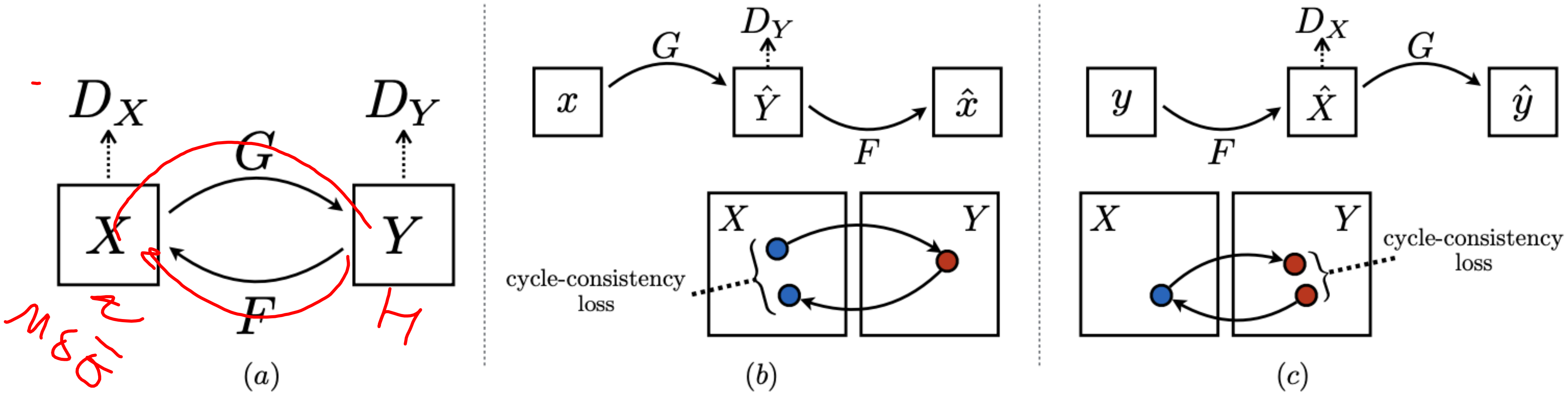


Figure 3: (a) Our model contains two mapping functions $G : X \to Y$ and $F : Y \to X$, and associated adversarial discriminators D_Y and D_X. D_Y encourages G to translate X into outputs indistinguishable from domain Y, and vice versa for D_X and F. To further regularize the mappings, we introduce two *cycle consistency losses* that capture the intuition that if we translate from one domain to the other and back again we should arrive at where we started: (b) forward cycle-consistency loss: $x \to G(x) \to F(G(x)) \approx x$, and (c) backward cycle-consistency loss: $y \to F(y) \to G(F(y)) \approx y$

https://arxiv.org/abs/1703.10593

Lab 3d

- Anomaly detection
- Discriminator
- No ground truth label?

TRAINING THE GAN

STEP 1

Input Journal Entry

duration	%%%%
src_bytes	%%%%
service	%%%%
flag	%%%%
src_bytes	%%%%
⋮	
dst_host_srv_rerror_rate	%%%%

Generate Fake Sample

Generator

Output Journal Entry

duration	#####
src_bytes	#####
service	#####
flag	#####
src_bytes	####
⋮	
dst_host_srv_rerror_rate	####

= MAE

Meaningless Output

TRAINING THE GAN

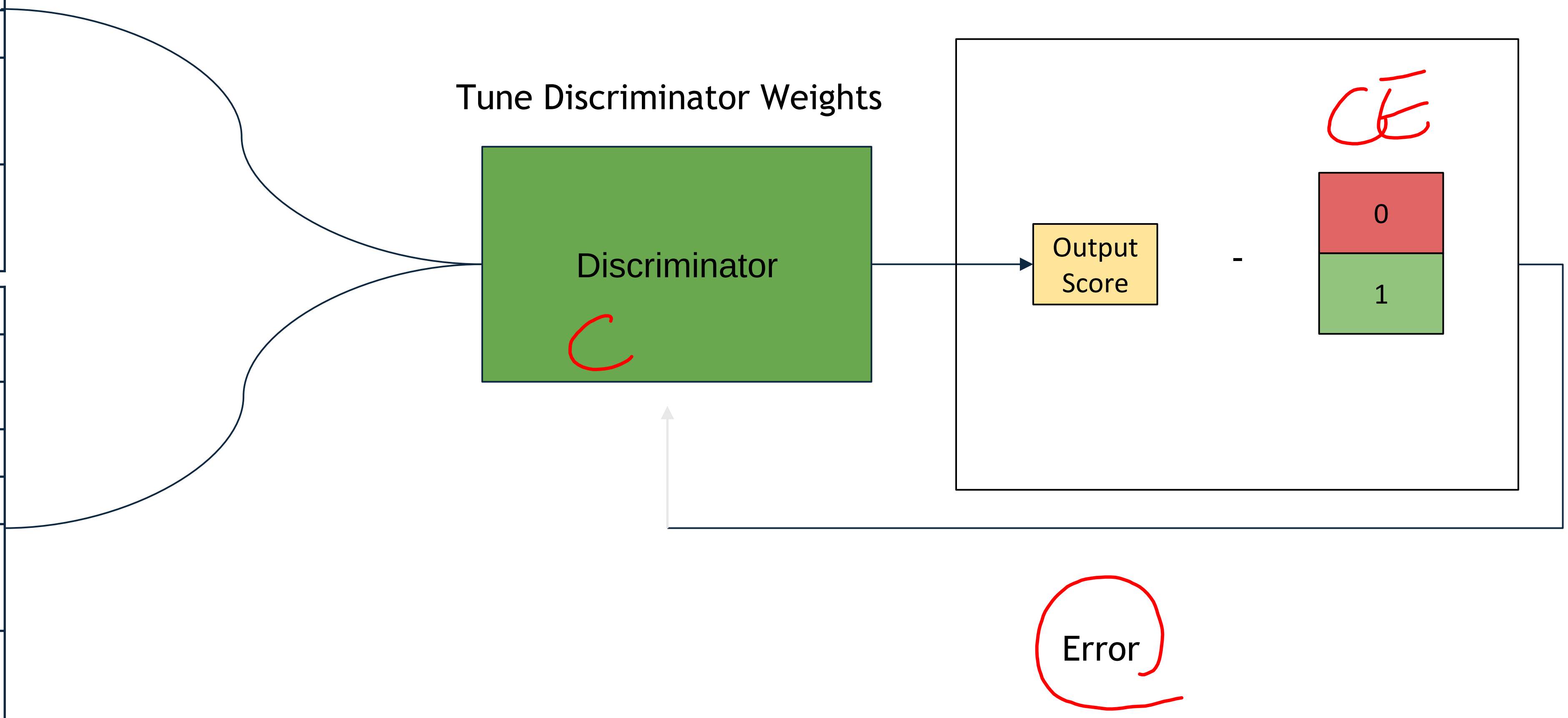
STEP 2

Generator Output

duration	#####
src_bytes	#####
service	#####
flag	#####
src_bytes	####
. . .	
dst_host_srv_rerror_rate	####

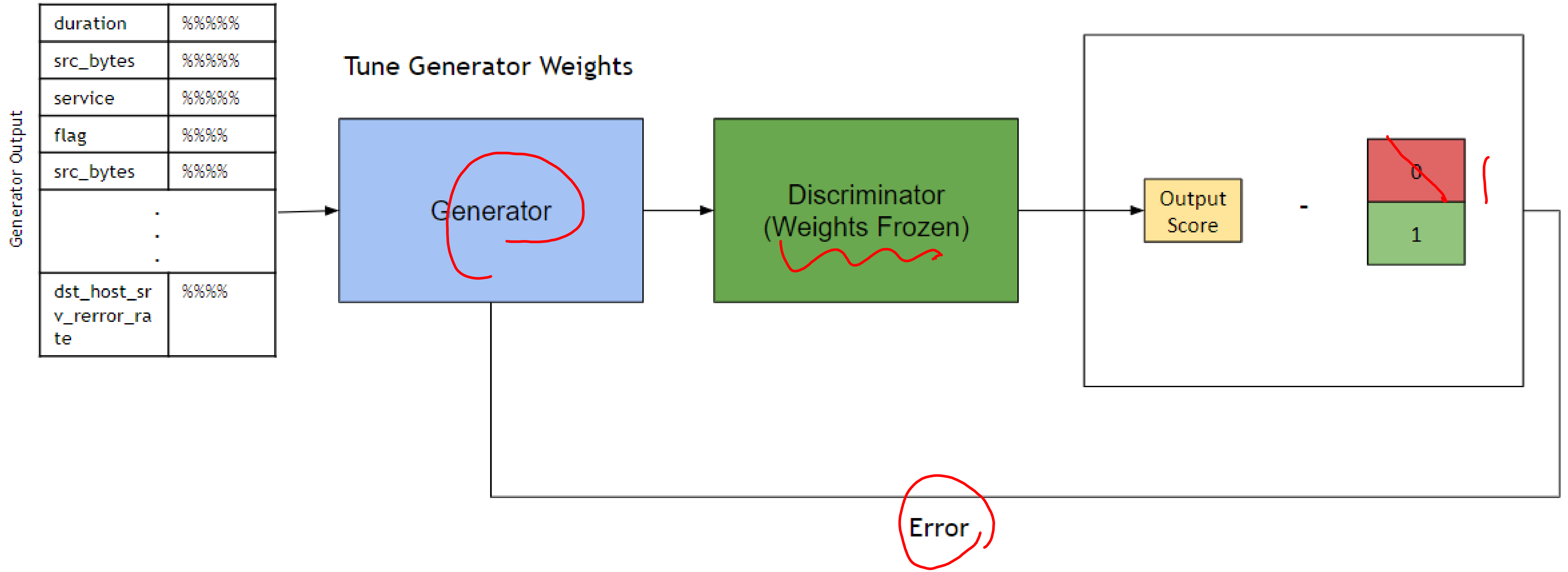
Real Data Sample

duration	0
src_bytes	215
service	http (1)
flag	SF (1)
src_bytes	215
. . .	
dst_host_srv_rerror_rate	0



TRAINING THE GAN

STEP 3



TRAINING THE GAN

BACK TO STEP 1

Input Journal Entry

duration	%%%%
src_bytes	%%%%
service	%%%%
flag	%%%%
src_bytes	%%%%
. . .	
dst_host_srv_rerror_rate	%%%%

Generate Fake Sample

Generator

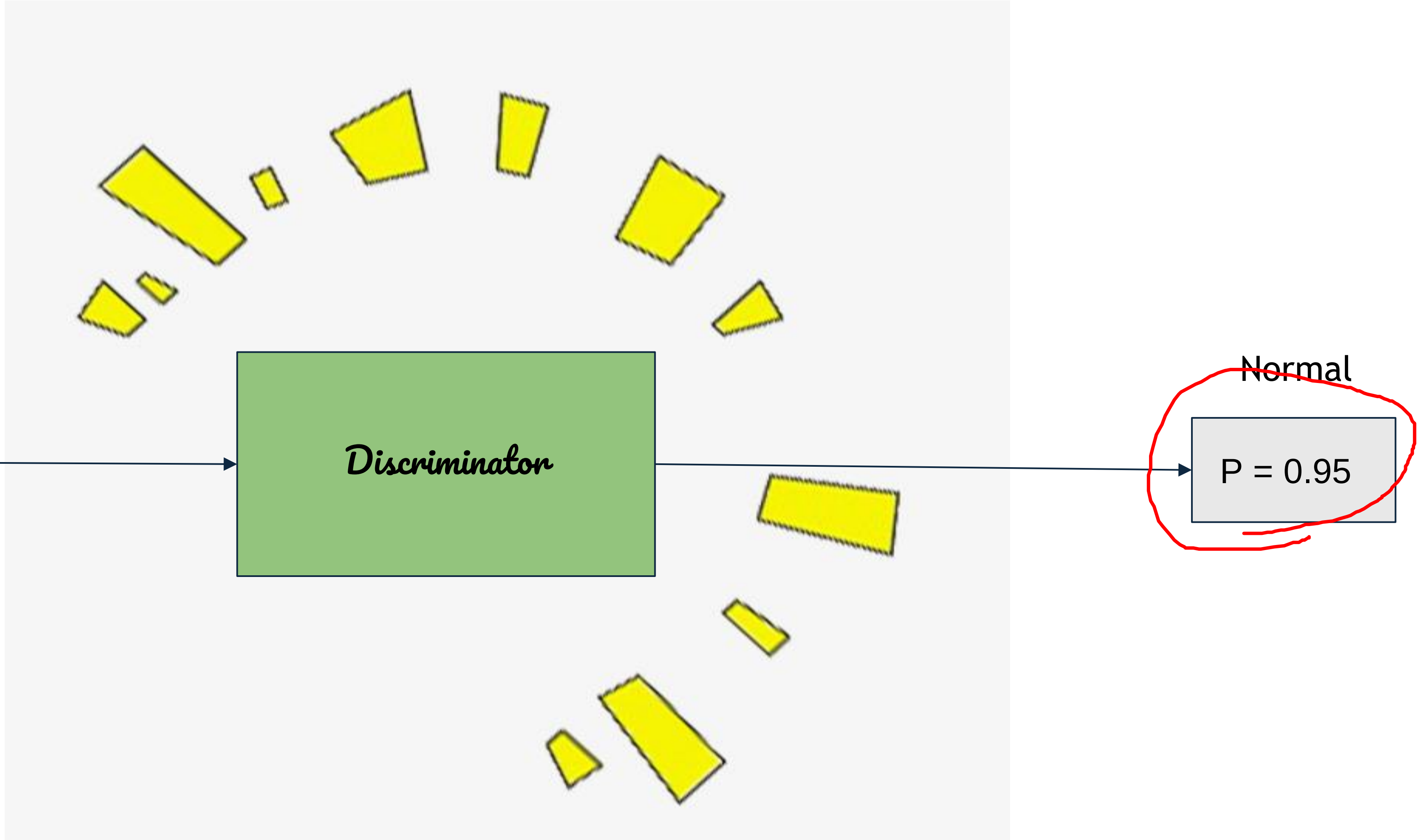
Output Journal Entry

duration	0
src_bytes	215
service	http (1)
flag	SF (1)
src_bytes	215
. . .	
dst_host_srv_rerror_rate	0

Meaningful Output

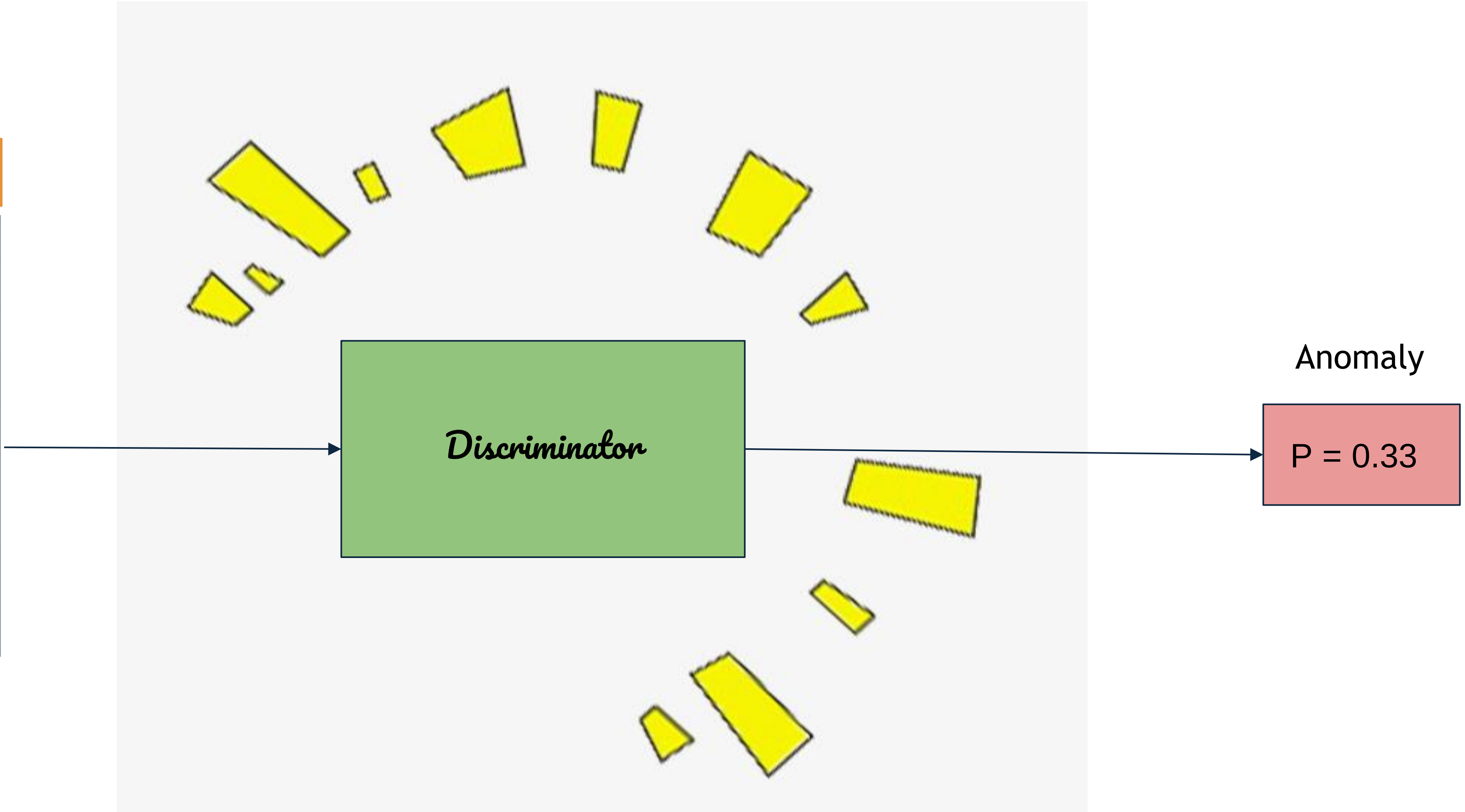
ANOMALY DETECTION

Input Test Point	
duration	0
src_bytes	215
service	http (1)
flag	SF (1)
src_bytes	215
⋮	
dst_host_srv_rerror_rate	0



ANOMALY DETECTION

Input Test point	
duration	99999
src_bytes	3239862
service	5454
flag	SF (1)
src_bytes	215
. . .	
dst_host_srv_rerror_rate	0



Thank You

Q&A